

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12403628

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Kobayashi; Akiyoshi et al.

Kneading method for use with kneading apparatus with first and second extruders

Abstract

A kneading apparatus includes a processor and an extruder. The extruder includes a screw. The screw includes a screw main body. A conveyance portion, a barrier portion, and a path are provided at places of the screw main body. In at least one of the places, the path is provided inside the screw main body and includes an entrance and an exit. The raw materials, pressure on which is increased by the barrier portion, flow in from the entrance. The raw materials flowing in from the entrance flow through the path toward the exit. The exit is positioned to be remote from the entrance in an axial direction. A method of using the kneading apparatus includes continuously melting and mixing materials in the processor, using the melted materials as the raw materials, and continuously discharging kneaded materials produced by kneading the raw materials with the screw in the extruder.

Inventors: Kobayashi; Akiyoshi (Tokyo, JP), Fujii; Shigeyuki (Numazu, JP), Sameshima; Takafumi (Mishima, JP), Shimizu; Hiroshi (Tsukuba, JP)

Applicant: SHIBAURA MACHINE CO., LTD. (Tokyo, JP); HSP Technologies Inc. (Ibaraki, JP)

Family ID: 54392469

Assignee: SHIBAURA MACHINE CO., LTD. (Tokyo, JP); HSP Technologies Inc. (Ibaraki, JP)

Appl. No.: 18/068753

Filed: December 20, 2022

Prior Publication Data

Document Identifier

US 20230125116 A1

Publication Date

Apr. 27, 2023

Foreign Application Priority Data

JP	2014-096800	May. 08, 2014
JP	2015-085231	Apr. 17, 2015

Related U.S. Application Data

continuation parent-doc WO PCT/JP2015/062549 20150424 PENDING child-doc US 15345711
division parent-doc US 15345711 20161108 US 11565440 child-doc US 18068753

Publication Classification

Int. Cl.: **B29B7/42** (20060101); **B29B7/48** (20060101); **B29B7/74** (20060101); **B29B7/82** (20060101); **B29B7/84** (20060101); **B29C48/385** (20190101); **B29C48/40** (20190101); **B29C48/51** (20190101); **B29C48/515** (20190101); **B29C48/57** (20190101); **B29C48/67** (20190101); **B29C48/76** (20190101)

U.S. Cl.:

CPC **B29B7/426** (20130101); **B29B7/429** (20130101); **B29B7/48** (20130101); **B29B7/483** (20130101); **B29B7/487** (20130101); **B29B7/489** (20130101); **B29B7/7461** (20130101); **B29B7/7485** (20130101); **B29B7/82** (20130101); **B29B7/84** (20130101); **B29B7/845** (20130101); **B29C48/385** (20190201); **B29C48/40** (20190201); **B29C48/402** (20190201); **B29C48/51** (20190201); **B29C48/515** (20190201); **B29C48/57** (20190201); **B29C48/67** (20190201); **B29C48/76** (20190201); **B29C48/767** (20190201);

Field of Classification Search

CPC: B29B (7/489); B29B (7/847); B29B (7/82); B29B (7/90); B29B (7/845); B29B (7/42); B29B (7/421); B29B (7/823); B29B (7/7485); B29B (7/7461); B29B (7/429); B29B (7/482); B29B (7/422); B29B (7/726); B29B (7/426); B29B (7/48); B29B (7/483); B29B (7/487); B29B (7/84); B29B (7/38); B29C (47/402); B29C (47/6018); B29C (47/622); B29C (47/6056); B29C (47/6012); B29C (48/82); B29C (48/802); B29C (48/625); B29C (48/385); B29C (48/535); B29C (48/55); B29C (48/54); B29C (48/72); B29C (48/402); B29C (48/74); B29C (48/745); B29C (48/725); B29C (48/375); B29C (48/40); B29C (48/51); B29C (48/515); B29C (48/57); B29C (48/67); B29C (48/76); B29C (48/767); B29K (2105/251); B29K (2023/12); B29K (2509/00)

USPC: 425/200-209; 366/79-91; 366/290; 366/75

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3305894	12/1966	Pfaff	366/85	B29C 48/268
3371379	12/1967	Reifenhauser	366/75	B29B 7/86
3503944	12/1969	Mohr	422/135	B29C 48/022

3712594	12/1972	Schippers	366/89	B29C 48/766
3746318	12/1972	Schippers	N/A	N/A
3799234	12/1973	Skidmore	528/502F	B29B 7/484
3924842	12/1974	Klein	366/318	B29C 48/725
3963558	12/1975	Skidmore	264/102	B29C 48/76
3999921	12/1975	Thor	366/76.9	B29B 7/42
4118163	12/1977	Lee	425/208	B29B 7/726
4169679	12/1978	Miller et al.	N/A	N/A
4290702	12/1980	Klein	366/77	B29B 7/429
4302409	12/1980	Miller et al.	N/A	N/A
4329313	12/1981	Miller et al.	N/A	N/A
4387997	12/1982	Klein	425/208	B29C 48/395
4423960	12/1983	Anders	366/85	B29C 48/40
4472059	12/1983	Klein	366/293	B29C 48/395
4600311	12/1985	Mourrier	366/322	B29C 48/505
4637790	12/1986	Klein	425/208	B29C 48/51
4802140	12/1988	Dowling	366/144	B29C 48/51
4902455	12/1989	Wobbe	264/102	B29B 7/726
4983114	12/1990	Hauck	366/85	B29C 48/40
5098267	12/1991	Cheng	366/76.9	B29C 44/3449
5102594	12/1991	Burlet	264/101	B29C 48/285
5358681	12/1993	Jerman et al.	N/A	N/A
5797677	12/1997	Weihrich	366/85	B29C 48/2564
5804111	12/1997	Kobayashi	425/149	B29C 48/57
6024479	12/1999	Haring	366/85	B29C 48/767
8048948	12/2010	Shimizu et al.	N/A	N/A
8975336	12/2014	Shimizu et al.	N/A	N/A
9199393	12/2014	Shimizu	N/A	B29B 7/28
10967554	12/2020	Kobayashi	N/A	B29B 7/483
11072104	12/2020	Kobayashi	N/A	B29B 7/845
11110638	12/2020	Kobayashi	N/A	B29C 48/57
11220022	12/2021	Kobayashi	N/A	B29C 48/535
11224991	12/2021	Kobayashi	N/A	B29B 7/7485
11229889	12/2021	Kobayashi	N/A	B29C 48/2564
11230033	12/2021	Kobayashi	N/A	B29B 7/42
11260468	12/2021	Frank	N/A	B23K 20/00
11400632	12/2021	Kobayashi	N/A	B29B 7/726
11400633	12/2021	Kobayashi	N/A	B29B 7/487

11565440	12/2022	Kobayashi	N/A	B29C 48/515
2004/0222543	12/2003	Innerebner et al.	N/A	N/A
2005/0087904	12/2004	Bryan	264/148	B29B 7/487
2006/0108706	12/2005	Galimberti	264/211.23	B29C 48/297
2016/0303766	12/2015	Kobayashi	N/A	N/A
2016/0332331	12/2015	Kobayashi	N/A	B29B 7/7485
2016/0332332	12/2015	Kobayashi	N/A	B29B 7/42
2017/0021547	12/2016	Kobayashi	N/A	B29C 48/298
2017/0050366	12/2016	Kobayashi	N/A	B29C 48/767
2017/0050367	12/2016	Kobayashi	N/A	B29B 7/483
2017/0113394	12/2016	Kobayashi	N/A	B29B 7/7461
2017/0225360	12/2016	Kobayashi	N/A	B29B 7/489
2017/0225379	12/2016	Kobayashi	N/A	B29C 48/49
2018/0093233	12/2017	Kobayashi	N/A	B29C 48/507
2018/0093234	12/2017	Kobayashi	N/A	B29B 7/483
2019/0352472	12/2018	Sameshima	N/A	B29C 48/022
2020/0282622	12/2019	Ougier	N/A	N/A
2021/0154906	12/2020	Kobayashi	N/A	B29B 7/7461
2021/0316492	12/2020	Kobayashi	N/A	B29B 7/845
2021/0354362	12/2020	Kobayashi	N/A	B29C 48/288
2021/0362374	12/2020	Sameshima	N/A	B29B 7/48
2022/0305704	12/2021	Sameshima	N/A	B29C 48/51
2022/0325455	12/2021	Sasai	N/A	D04H 1/4291
2023/0091065	12/2022	Chastek	N/A	B29C 48/252
2024/0033974	12/2023	Sasai	N/A	B29B 7/90

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
87102748	12/1986	CN	N/A
101973121	12/2010	CN	N/A
25 48 490	12/1976	DE	N/A
27 58 265	12/1981	DE	N/A
43 19 058	12/1993	DE	N/A
69417466	12/1998	DE	N/A
0 688 600	12/1994	EP	N/A
2 578 378	12/2012	EP	N/A
3650196	12/2019	EP	N/A

3650196	12/2019	EP	N/A
1 501 412	12/1977	GB	N/A
52-72573	12/1976	JP	N/A
S52-72573	12/1976	JP	N/A
56-037054	12/1980	JP	N/A
57-107826	12/1981	JP	N/A
59-184635	12/1983	JP	N/A
H04-044826	12/1991	JP	N/A
5-220818	12/1992	JP	N/A
07-088923	12/1994	JP	N/A
2005-169764	12/2004	JP	N/A
2010-069771	12/2009	JP	N/A
2011-046104	12/2010	JP	N/A
2012-051289	12/2011	JP	N/A
10-2010-0087738	12/2009	KR	N/A
889462	12/1980	SU	N/A
996222	12/1982	SU	N/A
WO 2010/061872	12/2009	WO	N/A
WO 2013-133453	12/2012	WO	N/A
WO 2015/163197	12/2014	WO	N/A
WO 2015/170617	12/2014	WO	N/A
WO 2020-025446	12/2019	WO	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 17/386,635, filed Sep. 28, 2023. cited by applicant

Japanese Office Action in JP Application No. 2015-082775 dated May 7, 2019. cited by applicant

German Office Action issued in DE Application No. 11 2015 002 161.2 dated Aug. 8, 2019. cited by applicant

International Search Report issued in International Application No. PCT/JP2015/060641 dated May 26, 2015. cited by applicant

International Search Report issued in International Application No. PCT/JP2015/061487 dated Jul. 7, 2015. cited by applicant

International Search Report issued in International Application No. PCT/JP2016/062278 dated Jul. 12, 2016. cited by applicant

International Search Report issued in International Application No. PCT/JP2016/062279 dated Jul. 12, 2016. cited by applicant

Written Opinion issued in International Application No. PCT/JP2015/060641 dated May 26, 2015. cited by applicant

Written Opinion issued in International Application No. PCT/JP2015/061487 dated Jul. 7, 2015. cited by applicant

Written Opinion issued in International Application No. PCT/JP2016/062278 dated Jul. 12, 2016. cited by applicant

Written Opinion issued in International Application No. PCT/JP2016/062279 dated Jul. 12, 2016. cited by applicant

Taiwan Office Action issued in Taiwan Application No. 104112230 dated Sep. 2, 2016. cited by applicant

Taiwan Office Action issued in Taiwan Application No. 104112709 dated Sep. 2, 2016. cited by applicant

Chinese Office Action issued in Chinese Application No. 201580007088.2 dated Mar. 3, 2017. cited by applicant

Chinese Office Action issued in Chinese Application No. 201580007825.9 dated Mar. 24, 2017. cited by applicant
Korean Office Action issued in KR 10-2016-7021148 dated Nov. 29, 2017. cited by applicant
English Language Abstract of JP 7-227836 published Aug. 29, 1995. cited by applicant
English Language Abstract of JP 2010-137405 published Jun. 24, 2010. cited by applicant
English Language Abstract of JP 2013-123841 published Jun. 24, 2013. cited by applicant
English Language Abstract of JP 2005-169764 published Jun. 30, 2005. cited by applicant
English Language Abstract of JP 57-041932 published Mar. 9, 1982. cited by applicant
English Language Abstract of KR 10-2010-0087738 published Aug. 5, 2010. cited by applicant
Office Action in CN Application No. 201680024052X dated Jul. 22, 2019. cited by applicant
Office Action in DE Application No. 112015001938.3 dated Jul. 26, 2019. cited by applicant
U.S. Appl. No. 15/221,293, filed Jul. 7, 2023. cited by applicant
U.S. Appl. No. 15/221,277, filed Jul. 7, 2023. cited by applicant
U.S. Appl. No. 15/795,534, filed Jul. 7, 2023. cited by applicant
U.S. Appl. No. 15/795,535, filed Jul. 7, 2023. cited by applicant
U.S. Appl. No. 15/345,750, filed Jul. 7, 2023. cited by applicant
U.S. Appl. No. 15/345,758, filed Jul. 7, 2023. cited by applicant
U.S. Appl. No. 17/166,781, filed Feb. 3, 2021 filed Jul. 7, 2023. cited by applicant
German Office Action in DE Application No. 11 2015 002 157.4, dated Sep. 27, 2021. cited by applicant
U.S. Appl. No. 15/345,711, filed Jul. 7, 2023. cited by applicant
U.S. Appl. No. 17/386,635, filed Jul. 7, 2023. cited by applicant
Office Action in German Application No. 112015002157.4, dated Aug. 13, 2024. cited by applicant

Primary Examiner: Cooley; Charles

Attorney, Agent or Firm: DLA Piper LLP (US)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional application of U.S. patent application Ser. No. 15/345,711, filed Nov. 8, 2016, and a Continuation Application of PCT Application No. PCT/JP2015/062549, filed Apr. 24, 2015 and based upon and claiming the benefit of priority from prior Japanese Patent Applications No. 2014-096800, filed May 8, 2014; and No. 2015-085231, filed Apr. 17, 2015, the entire contents of all of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a kneading apparatus comprising an extruder which kneads blended raw materials while imparting shearing action and extension action to them, and a kneading method thereof.

2. Description of the Related Art

(2) For example, if raw materials obtained by blending incompatible resins are kneaded by an extruder wherein the rotational rate of a screw is set at approximately 300 rpm, it is necessary to add a compatibilizer having affinity for or adhesive properties to one or both of the blended components. However, even if the compatibilizer is used, the blended components are not mutually melted at a molecular level. Thus, naturally, there is a limit on the improvement in the capabilities and the functions of kneaded materials produced by the extruder.

(3) To solve such a problem, there has been conventionally developed a high shearing and forming apparatus capable of producing kneaded materials having a microscopic structure in which if one macromolecular component is formed into a matrix, the dispersed phase size of the other macromolecular component is controlled to a diameter of 300 nm or less, or a structure in which both the macromolecular components are microscopically connected to each other, without using any additives such as a compatibilizer.

(4) The high shearing and forming apparatus disclosed in Patent Literature 1 comprises a feedback-type screw accommodated in a cylinder. The screw has a structure in which raw materials obtained by blending incompatible resins are sufficiently kneaded inside the screw.

(5) Specifically, the screw has a straight axial line in the conveyance direction of the raw materials, and is configured to rotate on the axial line inside the cylinder. A spiral flight which conveys raw materials supplied from one end portion of the screw in the axial direction of the screw is formed on the outer circumferential surface of the screw. The raw materials conveyed by the flight are trapped in a gap between a tip surface of the screw and a sealing member closing an opening end of the cylinder. Moreover, the screw has a hole having an inside diameter of approximately 1 to 5 mm substantially at its central portion. The hole extends in the axial-line direction of the screw. An upstream end of the hole is opened in the gap in the tip surface of the screw. A downstream end of the hole branches bifurcately, and is opened in the outer circumferential surface of the one end portion of the screw.

(6) Thus, the raw materials trapped in the gap flow into the hole from the upstream end of the hole with the rotation of the screw, and are returned to the outer circumferential surface of the one end portion of the screw from the downstream end of the hole. The returned raw materials are conveyed again toward the gap by the flight.

(7) In this manner, by using the feedback-type screw, raw materials supplied to the screw are subjected to shearing action in the process of being conveyed by the flight and subjected to extension action in the process of passing through the hole. As a result, the degree of kneading of raw materials is increased by high shearing. Thus, macromolecular components of the raw materials are dispersed at a nanolevel, and kneaded materials having a microscopic dispersion structure can be obtained.

CITATION LIST

Patent Literature

(8) Patent Literature 1

(9) WO 2010/061872 A

Patent Literature 2 JP 2011-020341 A

BRIEF SUMMARY OF THE INVENTION

Technical Problem

(10) Incidentally, the apparatus of Patent Literature 1 repeats a circulation process of feeding raw materials supplied to the cylinder from a rear end of the screw to a gap in a front end, and then returning them from the gap to the rear end of the screw. However, in the circulation process, the next kneading is performed after a particular amount of raw materials is circulated and kneaded and the kneaded materials are discharged. Accordingly, a particular amount or more of kneaded materials cannot be continuously discharged.

(11) Moreover, Patent Literature 2 discloses a kneader comprising a kneading portion which kneads a particular amount of materials to be kneaded, and a buffer portion in which the fluid of kneaded materials is collected. A particular amount of kneaded fluid is supplied from the kneading portion to the buffer portion, and collected. The collected fluid is continuously discharged from the buffer portion.

(12) In the kneader of Patent Literature 2, in a state in which a predetermined amount of the fluid of kneaded materials is collected in advance in the buffer portion, the discharge of the fluid from the buffer portion is controlled so that the collected fluid is not depleted. That is, the amount of fluid

discharged from the buffer portion is limited. Thus, the fluid is continuously discharged within the range of the residual amount of fluid currently collected in the buffer portion until fluid is next supplied from the kneading portion.

(13) However, the discharge control of Patent Literature 2 is exercised to temporarily collect the fluid intermittently supplied from the kneading portion in the buffer portion, and then discharge the fluid in a regular small amount within the range of the collected amount, not to continuously convey the fluid in the process of being kneaded and discharged. In other words, the discharge control of Patent Literature 2 is merely exercised as if the fluid were continuously discharged. Because a process whereby the fluid stagnates in the process of being kneaded and discharged is essential, the fluid cannot be completely continuously produced. As a result, the kneader of Patent Literature 2 cannot continuously discharge a particular amount or more of fluid.

(14) Moreover, in the kneader of Patent Literature 2, the fluid of kneaded materials is temporarily collected in the buffer portion. Thus, the physical properties of the fluid collected in the buffer portion may change depending on the length of time it spends in the buffer portion.

(15) To maintain uniform quality, it is required that continuously discharged fluid has the same physical properties over its total length or whole. Nevertheless, if fluid whose physical properties have changed is mixed, portions of different physical properties exist in the fluid. Thus, the quality of the fluid to be discharged cannot be uniformly maintained.

(16) Accordingly, an object of the present invention is to provide to a kneading technique of continuously conveying processing objects without their stagnating in all the processes including a melt process for materials, a kneading process in which melted materials are used as raw materials, and a discharge process for produced kneaded materials, and thereby enabling completely continuous production of kneaded materials having uniform quality.

Solution to Problem

(17) In general, according to one embodiment, a kneading apparatus includes a processor, and an extruder. The extruder includes a screw. The screw includes a screw main body. A conveyance portion, a barrier portion, and a path are provided at places of the screw main body. In at least one of the places, the path is provided inside the screw main body, and includes an entrance and an exit. The raw materials, pressure on which is increased by the barrier portion, flow in from the entrance. The raw materials flowing in from the entrance flow through the path toward the exit. The exit is positioned to be remote from the entrance in an axial direction.

(18) In general, according to one embodiment, a kneading method comprising: continuously melting and mixing materials in a processor; and using the melted materials as raw materials and continuously discharging kneaded materials produced by kneading the raw materials with a screw in an extruder, the screw comprising a screw main body configured to rotate on a straight axial line in a conveyance direction of the raw materials, a path through which the raw materials flow being provided inside the screw main body, wherein in the extruder, the raw materials conveyed along an outer circumferential surface of the screw main body flow through the path and then return to an outer circumferential surface of the screw, while the kneaded materials are continuously discharged.

Advantageous Effects of Invention

(19) According to the present invention, a kneading technique of continuously conveying processing objects without their stagnating in all the processes including a melt process for materials, a kneading process in which melted materials are used as raw materials, and a discharge process for produced kneaded materials, and thereby enabling completely continuous production of kneaded materials having uniform quality.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING

- (1) FIG. 1 is a perspective view schematically showing a continuous high shearing processing apparatus (kneading apparatus) according to a first embodiment.
- (2) FIG. 2 is a sectional view of a first extruder used in the first embodiment.
- (3) FIG. 3 is a perspective view showing a state in which two screws of the first extruder engage with each other in the first embodiment.
- (4) FIG. 4 is a sectional view of a third extruder used in the first embodiment.
- (5) FIG. 5 is a sectional view of a second extruder used in the first embodiment.
- (6) FIG. 6 is a sectional view of the second extruder sectionally showing both a barrel and a screw in the first embodiment.
- (7) FIG. 7 is a side view of the screw used in the first embodiment.
- (8) FIG. 8 is a sectional view along line F8-F8 of FIG. 6.
- (9) FIG. 9 is a sectional view along line F9-F9 of FIG. 6.
- (10) FIG. 10 is a side view showing a flow direction of raw materials with respect to the screw in the first embodiment.
- (11) FIG. 11 is a sectional view of the second extruder schematically showing the flow direction of raw materials when the screw rotates in the first embodiment.
- (12) FIG. 12 is a sectional view of a second extruder used in a second embodiment.
- (13) FIG. 13 is a sectional view of a second extruder used in a third embodiment.
- (14) FIG. 14 is a sectional view of the second extruder sectionally showing both a barrel and a screw in the third embodiment.
- (15) FIG. 15 is a sectional view along line F15-F15 of FIG. 14.
- (16) FIG. 16 is a perspective view of a tube used in the third embodiment.
- (17) FIG. 17 is a side view showing a flow direction of raw materials with respect to the screw in the third embodiment.
- (18) FIG. 18 is a sectional view of the second extruder schematically showing the flow direction of raw materials when the screw rotates in the third embodiment.
- (19) FIG. 19 is a sectional view of a second extruder used in a fourth embodiment.
- (20) FIG. 20 is a sectional view of the second extruder sectionally showing both a barrel and a screw in the fourth embodiment.
- (21) FIG. 21 is a side view of the screw used in the fourth embodiment.
- (22) FIG. 22 is a sectional view along line F22-F22 of FIG. 20.
- (23) FIG. 23 is a perspective view of a tube used in the fourth embodiment.
- (24) FIG. 24 is a transverse sectional view of the tube shown in FIG. 23.
- (25) FIG. 25 is a perspective view showing another structural example of the tube used in the fourth embodiment.
- (26) FIG. 26 is a side view showing a flow direction of raw materials with respect to the screw in the fourth embodiment.
- (27) FIG. 27 is a sectional view of the second extruder schematically showing the flow direction of raw materials when the screw rotates in the fourth embodiment.
- (28) FIG. 28 is a sectional view of a first extruder according to a fifth embodiment.
- (29) FIG. 29 is a sectional view of a second extruder according to a sixth embodiment.
- (30) FIG. 30 is a sectional view of a second extruder according to a seventh embodiment.
- (31) FIG. 31 is a sectional view of a second extruder according to an eighth embodiment.
- (32) FIG. 32 is a sectional view of a second extruder according to a ninth embodiment.

DETAILED DESCRIPTION OF THE INVENTION

First Embodiment

- (33) A first embodiment will be described hereinafter with reference to FIG. 1 to FIG. 11.
- (34) FIG. 1 schematically shows a structure of a continuous high shearing processing apparatus

(kneading apparatus) **1** according to the first embodiment. The high shearing processing apparatus **1** comprises a first extruder (processor) **2**, a second extruder **3**, and a third extruder (deaerator) **4**. The first extruder **2**, the second extruder **3**, and the third extruder **4** are connected to each other in series.

(35) The first extruder **2** is a processor for preliminarily kneading and melting materials, for example, two kinds of incompatible resin. Here, a polycarbonate (PC) resin and a polymethyl methacrylate (PMMA) resin are applied as the two kinds of resin. These resins are supplied to the first extruder **2**, for example, in a state of pellets.

(36) In the present embodiment, a corotating twin screw kneader is used as the first extruder **2** to increase the degree of kneading and melting of the resins. FIG. 2 and FIG. 3 show an example of the twin screw kneader. The twin screw kneader comprises a barrel **6** and two screws **7a** and **7b** accommodated inside the barrel **6**. The barrel **6** includes a cylinder portion **8** having the shape of a combination of two cylinders. The resins are continuously supplied to the cylinder portion **8** from a supply port **9** provided at one end portion of the barrel **6**. Moreover, the barrel **6** contains a heater for melting the resins.

(37) The screws **7a** and **7b** are accommodated in the cylinder portion **8** in a state of engaging with each other. The screws **7a** and **7b** are rotated in the same direction upon receipt of torque transmitted from a motor not shown in the figures. As shown in FIG. 3, the screws **7a** and **7b** each comprise a feed portion **11**, a kneading portion **12**, and a pumping portion **13**. The feed portion **11**, the kneading portion **12**, and the pumping portion **13** are arranged in a line in the axial direction of the screws **7a** and **7b**.

(38) The feed portion **11** comprises a spirally twisted flight **14**. The flights **14** of the screws **7a** and **7b** rotate in a state of engaging with each other, and convey two kinds of resin supplied from the supply port **9** toward the kneading portion **12**.

(39) The kneading portion **12** comprises disks **15** arranged in the axial direction of the screws **7a** and **7b**. The disks **15** of the screws **7a** and **7b** rotate in a state of facing each other, and preliminarily knead resin fed from the feed portion **11**. The kneaded resin is fed to the pumping portion **13** by the rotation of the screws **7a** and **7b**.

(40) The pumping portion **13** comprises a spirally twisted flight **16**. The flights **16** of the screws **7a** and **7b** rotate in a state of engaging each other, and extrude preliminarily kneaded resin from a discharge end of the barrel **6**.

(41) According to the above-described twin screw kneader, resin supplied to the feed portion **11** of the screws **7a** and **7b** is melted by shearing heat produced by the rotation of the screws **7a** and **7b** and heat of the heater. The resin melted by being preliminarily kneaded by the twin screw kneader constitutes blended raw materials. As indicated by arrow A in FIG. 1, the raw materials are continuously supplied from the discharge end of the barrel **6** to the second extruder **3**.

(42) Moreover, since the first extruder **2** is formed as a twin screw kneader, it is possible, not only to melt resin, but also to impart shearing action to the resin. Thus, at a point in time when raw materials are supplied to the second extruder **3**, the raw materials are melted by being preliminarily kneaded by the first extruder **2** and held at optimum viscosity. In addition, since the first extruder **2** is formed as a twin screw kneader, a predetermined amount of raw materials can be stably supplied per unit time when raw materials are continuously supplied to the second extruder **3**. Accordingly, the workload of the second extruder **3**, which kneads raw materials thoroughly, can be reduced.

(43) The second extruder **3** is an element for producing kneaded materials having a microscopic dispersion structure in which macromolecular components of raw materials are dispersed at a nanolevel. In the present embodiment, a single screw extruder is used as the second extruder **3**. The single screw extruder comprises a barrel **20** and a screw **21**. The screw **21** has the function of repeatedly imparting shearing action and extension action to melted raw materials. The structure of the second extruder **3** including the screw **21** will be described later in detail.

(44) The third extruder **4** is an element for drawing and removing gaseous components included in

kneaded materials discharged from the second extruder 3. In the present embodiment, a single screw extruder is used as the third extruder 4. As shown in FIG. 4, the single screw extruder comprises a barrel 22 and a vented screw 23 accommodated in the barrel 22. The barrel 22 includes a cylinder portion 24 having a straight cylindrical shape. Kneaded materials extruded from the second extruder 3 are continuously supplied to the cylinder portion 24 from one end portion in the axial direction of the cylinder portion 24.

(45) The barrel 22 comprises a vent-port 25. The vent-port 25 is opened in a middle portion in the axial direction of the cylinder portion 24, and is connected to a vacuum pump 26. Moreover, the other end portion of the cylinder portion 24 of the barrel 22 is closed by a head portion 27. The head portion 27 comprises a discharge port 28 through which kneaded materials are discharged.

(46) The vented screw 23 is accommodated in the cylinder portion 24. The vented screw 23 is rotated in one direction upon receipt of torque transmitted from a motor not shown in the figures. The vented screw 23 comprises a spirally twisted flight 29. The flight 29 rotates integrally with the vented screw 23, and continuously conveys kneaded materials supplied to the cylinder portion 24 toward the head portion 27. The kneaded materials are subjected to vacuum pressure of the vacuum pump 26 when being conveyed to a position corresponding to the vent-port 25. That is, gaseous materials and other volatile components included in kneaded materials are continuously drawn and removed from the kneaded materials by creating negative pressure in the cylinder portion 24 by the vacuum pump. The kneaded materials, from which gaseous materials and other volatile components are removed, are continuously discharged from the discharge port 28 of the head portion 27 to the outside of the high shearing processing apparatus 1.

(47) Next, the second extruder 3 will be described in detail.

(48) As shown in FIG. 5 and FIG. 6, the barrel 20 of the second extruder 3 has a straight tubular shape, and is horizontally disposed. The barrel 20 is divided into barrel elements 31.

(49) Each of the barrel elements 31 comprises a cylindrical through-hole 32. The barrel elements 31 are integrally joined, such that the respective through-holes 32 coaxially continue. The through-holes 32 of the barrel elements 31 define a cylinder portion 33 having a cylindrical shape inside the barrel 20 in cooperation with each other. The cylinder portion 33 extends in the axial direction of the barrel 20.

(50) A supply port 34 is formed in one end portion in the axial direction of the barrel 20. The supply port 34 communicates with the cylinder portion 33, and raw materials blended by the first extruder 2 are continuously supplied to the supply port 34.

(51) The barrel 20 comprises a heater not shown in the figures. The heater adjusts the temperature of the barrel 20 to an optimum value for kneading raw materials. Moreover, the barrel 20 comprises refrigerant paths 35 through which a refrigerant, for example, water or oil, flows. The refrigerant paths 35 are disposed to surround the cylinder portion 33. The refrigerant flows along the refrigerant paths 35 and forcibly refrigerates the barrel 20, when the temperature of the barrel 20 exceeds a predetermined upper limit.

(52) The other end portion in the axial direction of the barrel 20 is closed by a head portion 36. The head portion 36 comprises a discharge port 36a. The discharge port 36a is positioned on the opposite side to the supply port 34 in the axial direction of the barrel 20, and is connected to the third extruder 4.

(53) As shown in FIG. 5 to FIG. 7, the screw 21 comprises a screw main body 37. The screw main body 37 of the present embodiment is composed of an axis of rotation 38 and cylindrical tubes 39.

(54) The axis of rotation 38 comprises a first axial portion 40 and a second axial portion 41. The first axial portion 40 is positioned at the basal end of the axis of rotation 38, which is the one end portion side of the barrel 20. The first axial portion 40 includes a joint portion 42 and a stopper portion 43. The joint portion 42 is coupled to a driving source such as a motor through a coupling not shown in the figures. The stopper portion 43 is coaxially provided on the joint portion 42. The stopper portion 43 is greater in diameter than the joint portion 42.

(55) The second axial portion **41** coaxially extends from an end face of the stopper portion **43** of the first axial portion **40**. The second axial portion **41** has a length substantially the same as the total length of the barrel **20**, and comprises a tip facing the head portion **36**. A straight axial line **O1** coaxially penetrating the first axial portion **40** and the second axial portion **41** horizontally extends in the axial direction of the axis of rotation **38**.

(56) The second axial portion **41** has a solid columnar shape smaller in diameter than the stopper portion **43**. As shown in FIG. **8** and FIG. **9**, a pair of keys **45a** and **45b** is attached to the outer circumferential surface of the second axial portion **41**. The keys **45a** and **45b** extend in the axial direction of the second axial portion **41** at positions shifted 180° in the circumferential direction of the second axial portion **41**.

(57) As shown in FIG. **6** to FIG. **9**, each of the tubes **39** is formed, such that the second axial portion **41** coaxially penetrates therethrough. A pair of keyways **49a** and **49b** is formed in the inner circumferential surfaces of the tubes **39**. The keyways **49a** and **49b** extend in the axial direction of the tubes **39** at positions shifted 180° in the circumferential direction of the tubes **39**.

(58) The tubes **39** are inserted on the second axial portion **41** from the direction of the tip of the second axial portion **41** in a state in which the keyways **49a** and **49b** are fitted to the keys **45a** and **45b** of the second axial portion **41**. In the present embodiment, a first collar **44** exists between the tube **39** inserted first on the second axial portion **41** and the end face of the stopper portion **43** of the first axial portion **40**. Moreover, after all the tubes **39** are inserted on the second axial portion **41**, a fixing screw **52** is screwed into a tip surface of the second axial portion **41** through a second collar **51**.

(59) All the tubes **39** are thereby constricted in the axial direction of the second axial portion **41** between the first collar **44** and the second collar **51**, and the end faces of the adjacent tubes **39** are firmly affixed to each other without any gap.

(60) At this time, all the tubes **39** are coaxially joined on the second axial portion **41**, whereby each of the tubes **39** and the axis of rotation **38** are integrally assembled. This makes it possible to rotate each of the tubes **39** on the axial line **O1** together with the axis of rotation **38**, that is, rotate the screw main body **37** on the axial line **O1**.

(61) In such a state, each of the tubes **39** serves as a structural element that defines the outside diameter **D1** (see FIG. **8**) of the screw main body **37**. That is, the outside diameters **D1** of the tubes **39** coaxially joined along the second axial portion **41** are set to be equal to each other. The outside diameter **D1** of the screw main body **37** (each of the tubes **39**) is defined as a diameter passing through the axial line **O1**, which is the center of rotation of the axis of rotation **38**.

(62) The segmental screw **21** in which the outside diameter **D1** of the screw main body **37** (each of the tubes **39**) is a fixed value is thereby formed. In the segmental screw **21**, screw elements can be held in free order and combination along the axis of rotation **38** (that is, the second axial portion **41**). With respect to the screw elements, for example, each of the tubes **39** on which at least parts of flights **56**, **57**, and **58**, which will be described later, are formed can be defined as one screw element.

(63) In this manner, by segmenting the screw **21**, its convenience can be significantly improved with respect to, for example, changes and adjustments to the specifications or the upkeep and maintenance of the screw **21**.

(64) In the present embodiment, the cylindrical tubes **39** are not restricted to being fixed to the axis of rotation **38** by the keys **45a** and **45b**. For example, the tubes **39** may be fixed to the axis of rotation **38** by a spline as shown in FIG. **2** instead of the keys **45a** and **45b**.

(65) Moreover, the segmental screw **21** is coaxially accommodated in the cylinder portion **33** of the barrel **20**. Specifically, the screw main body **37** with the screw elements held along the axis of rotation **38** (the second axial portion **41**) is rotatably accommodated in the cylinder portion **33**. In this state, the first axial portion **40** (the joint portion **42** and the stopper portion **43**) of the axis of rotation **38** projects from the one end portion of the barrel **20** to the outside of the barrel **20**.

(66) Moreover, in this state, a conveyance path 53 for conveying raw materials is formed between the outer circumferential surface in the circumferential direction of the screw main body 37 and the inner circumferential surface of the cylinder portion 33. The conveyance path 53 has an annular sectional shape in the radial direction of the cylinder portion 33, and extends in the axial direction along the cylinder portion 33.

(67) In the present embodiment, the screw 21 rotates left-handed in an anticlockwise direction from the perspective of the basal end side of the screw 21 as indicated by an arrow in FIG. 5 upon receipt of torque from a driving source. At this time, it is preferable that the rotational rate of the screw 21 be 600 to 3,000 rpm.

(68) As shown in FIG. 5 to FIG. 7, the screw main body 37 comprises conveyance portions 54 and 59 for conveying raw materials and barrier portions 55 for limiting the flow of raw materials. To be specific, a barrier portion 55 is disposed at the basal end of the screw main body 37, which corresponds to the one end portion of the barrel 20, and the discharge conveyance portion 59 is disposed at the tip of the screw main body 37, which corresponds to the other end portion of the barrel 20. Moreover, between the barrier portion 55 and the conveyance portion 59, the conveyance portions 54 and the barrier portions 55 are alternately disposed in the axial direction from the basal end toward the tip of the screw main body 37.

(69) The supply port 34 of the barrel 20 opens toward the conveyance portion 54 disposed on the basal end side of the screw main body 37.

(70) Each of the conveyance portions 54 comprises the flight 56 spirally twisted. The flight 56 projects from the outer circumferential surface in the circumferential direction of the tubes 39 toward the conveyance path 53. The flight 56 is twisted to convey raw materials from the tip toward the basal end of the screw main body 37 when the screw 21 rotates left-handed. That is, the flight 56 is twisted left-handed as in the case of a left-handed screw.

(71) Moreover, the discharge conveyance portion 59 comprises the flight 58 spirally twisted. The flight 58 projects from the outer circumferential surface in the circumferential direction of the tubes 39 toward the conveyance path 53. The flight 58 is twisted to convey raw materials from the basal end toward the tip of the screw main body 37 when the screw 21 rotates left-handed. That is, the flight 58 is twisted right-handed as in the case of a right-handed screw.

(72) Each of the barrier portions 55 comprises the flight 57 spirally twisted. The flight 57 projects from the outer circumferential surface in the circumferential direction of the tubes 39 toward the conveyance path 53. The flight 57 is twisted to convey raw materials from the basal end toward the tip of the screw main body 37 when the screw 21 rotates left-handed. That is, the flight 57 is twisted right-handed as in the case of a right-handed screw.

(73) The twist pitch of the flight 57 of each of the barrier portions 55 is set to be smaller than or equal to those of the flights 56 and 58 of the conveyance portions 54 and 59. Moreover, a slight clearance is secured between the apexes of the flights 56, 57, and 58 and the inner circumferential surface of the cylinder portion 33 of the barrel 20.

(74) In this case, the clearance between the outside diameter portions of the barrier portions 55 (the apexes of the flights 57) and the inner circumferential surface of the cylinder portion 33 is preferably set to be within a range of 0.1 to 2 mm. More preferably, the clearance is set to be within a range of 0.1 to 0.7 mm. This can surely limit the conveyance of raw materials through the clearance.

(75) The axial direction of the screw main body 37 can also be referred to as the longitudinal direction of the screw main body 37, in other words, the longitudinal direction of the screw 21.

(76) Here, the lengths of the conveyance portions 54 and 59 in the axial direction of the screw main body 37 are set as appropriate in accordance with, for example, the kind of raw materials, the degree of kneading of raw materials, and the production of kneaded materials per unit time. The conveyance portions 54 and 59 are at least areas where the flights 56 and 58 are formed on the outer circumferential surfaces of the tubes 39, but are not limited to areas between the start points

and the end points of the flights **56** and **58**.

(77) That is, areas outside the flights **56** and **58** of the outer circumferential surfaces of the tubes **39** may be regarded as the conveyance portions **54** and **59**. For example, if a cylindrical spacer or a cylindrical collar is disposed at a position adjacent to the tubes **39** comprising the flights **56** and **58**, the spacer or the collar also can be included in the conveyance portions **54** and **59**.

(78) Moreover, the lengths of the barrier portions **55** in the axial direction of the screw main body **37** are set as appropriate in accordance with, for example, the kind of raw materials, the degree of kneading of raw materials, and the production of kneaded materials per unit time. The barrier portions **55** function to stop the flow of raw materials fed by the conveyance portions **54**. That is, the barrier portions **55** are adjacent to the conveyance portions **54** on the downstream side in the conveyance direction of raw materials, and configured to prevent raw materials fed by the conveyance portions **54** from passing through the clearance between the apexes of the flights **57** and the inner circumferential surface of the cylinder portion **33**.

(79) Moreover, in the above-described screw **21**, each of the flights **56**, **57**, and **58** projects from the outer circumferential surfaces of the tubes **39** having the outside diameters **D1** equal to each other toward the conveyance path **53**. Thus, the outer circumferential surface in the circumferential direction of each of the tubes **39** defines the root diameter of the screw **21**. The root diameter of the screw **21** is kept at a fixed value over the total length of the screw **21**.

(80) As shown in FIG. 5 to FIG. 7 and FIG. 10, the screw main body **37** comprises paths **60** extending in the axial direction inside the screw main body **37**. If one barrier portion **55** and two conveyance portions **54**, between which the barrier portion **55** is sandwiched, are regarded as one unit, a path **60** is formed over tubes **39** of the pair of conveyance portions **54** and a tube **39** of the barrier portion **55**.

(81) In this case, the paths **60** are arranged at predetermined intervals (for example, regular intervals) in the axial direction of the screw main body **37**. In addition, in a middle portion in the axial direction of the screw main body **37**, four paths **60** extending in the axial direction of the screw main body **37** are arranged at intervals of 90° in the circumferential direction of the screw main body **37**.

(82) Moreover, the paths **60** are provided at positions eccentric to the axial line **O1** of the axis of rotation **38** inside the tubes **39**. In other words, the paths **60** are shifted from the axial line **O1**, and revolve around the axial line **O1** when the screw main body **37** rotates.

(83) As shown in FIG. 8 and FIG. 9, the paths **60** are, for example, holes having a circular sectional shape. The inside diameter of the holes is, for example, set to be greater than or equal to 1 mm but less than 6 mm, and preferably, greater than or equal to 1 mm but less than 5 mm. Moreover, the tubes **39** of the conveyance portions **54** and the barrier portions **55** comprise tubular wall surfaces **61** defining the holes. That is, the paths **60** are holes composed of hollow spaces only, and the wall surfaces **61** continuously surround the hollow paths **60** in the circumferential direction. The paths **60** are thereby formed as hollow spaces which allow only the flow of raw materials. In other words, inside the paths **60**, there are no other elements constituting the screw main body **37**. Moreover, the wall surfaces **61** revolve around the axial line **O1** without rotating on the axial line **O1**, when the screw main body **37** rotates.

(84) As shown in FIG. 5, FIG. 6, and FIG. 11, each of the paths **60** comprises an entrance **62**, an exit **63**, and a path main body **64** connecting the entrance **62** and the exit **63**. The entrance **62** and the exit **63** are provided to be remote from both sides of one barrier portion **55**. Specifically, in the conveyance portion **54** adjacent to the barrier portion **55** from the basal end side of the screw main body **37**, the entrance **62** is opened in the outer circumferential surface near the downstream end of the conveyance portion **54**. In addition, in the conveyance portion **54** adjacent to the barrier portion **55** from the tip side of the screw main body **37**, the exit **63** is opened in the outer circumferential surface near the upstream end of the conveyance portion **54**.

(85) The path main body **64** extends straight without branching on the way in the axial direction of

the screw main body **37**. The figures show, as an example, a state in which the path main body **64** extends parallel to the axial line **O1**. Both sides of the path main body **64** are closed in the axial direction.

(86) The entrance **62** is provided on one side of the path main body **64**, that is, a portion closer to the basal end of the screw main body **37**. In this case, the entrance **62** may be opened in the outer circumferential surface of the screw main body **37** from an end face on the one side of the path main body **64**, or may be opened in the outer circumferential surface of the screw main body **37** from a portion closer to the end face on the one side of the path main body **64**, that is, a portion located short of the end face. The opening direction of the entrance **62** is not limited to those orthogonal to the axial line **O1**, but may be those crossing the axial line **O1**. In this case, entrances **62** may be provided by opening one side of the path main body **64** in directions.

(87) From another point of view, the entrance **62** is opened in the outer circumferential surface of the conveyance portion **54** more remote in the direction of the basal end of the screw main body **37** than the barrier portion **55** in the above-described one unit. It is preferable that the entrance **62** be provided at a position most remote in the direction of the basal end of the screw main body **37** on the outer circumferential surface of the tube **39** constituting the conveyance portion **54**. The entrance **62** is thereby positioned just before the barrier portion **55** adjacent to the conveyance portion **54** in which the entrance **62** is opened in the direction of the basal end of the screw main body **37**.

(88) The exit **63** is provided on the other side (the opposite side to the one side) of the path main body **64**, that is, a portion closer to the tip of the screw main body **37**. In this case, the exit **63** may be opened in the outer circumferential surface of the screw main body **37** from an end face on the other side of the path main body **64**, or may be opened in the outer circumferential surface of the screw main body **37** from a portion closer to the end face on the other side of the path main body **64**, that is, a portion located short of the end face. The opening direction of the exit **63** is not limited to those orthogonal to the axial line **O1**, but may be those crossing the axial line **O1**. In this case, exits **63** may be provided by opening one side of the path main body **64** in directions.

(89) From another point of view, the exit **63** is opened in the outer circumferential surface of the conveyance portion **54** more remote in the direction of the tip of the screw main body **37** than the barrier portion **55** in the above-described one unit. It is preferable that the exit **63** be provided at a position most remote in the direction of the tip of the screw main body **37** on the outer circumferential surface of the tube **39** constituting the conveyance portion **54**. The exit **63** is thereby positioned just before the barrier portion **55** adjacent to the conveyance portion **54** in which the exit **63** is opened in the direction of the tip of the screw main body **37**.

(90) The path main body **64** connecting the entrance **62** and the exit **63** traverses the barrier portion **55** of the above-described one unit, and has a length stretching between the two conveyance portions **54**, between which the barrier portion **55** is sandwiched. In this case, the bore of the path main body **64** may be set to be smaller than those of the entrance **62** and the exit **63**, or may be set to be equal to them. In either case, the path sectional area defined by the bore of the path main body **64** is set to be much smaller than the annular sectional area in the radial direction of the above-described annular conveyance path **53**.

(91) In the present embodiment, if the screw **21** is disassembled by detaching the tubes **39** on which the flights **56**, **57**, and **58** are formed from the axis of rotation **38**, the tubes **39** on which at least parts of the flights **56**, **57**, and **58** are formed can also be referred to as the above-described screw elements.

(92) Thus, the screw main body **37** of the screw **21** can be formed by sequentially inserting the tubes **39** as the screw elements on the outer circumference of the axis of rotation **38**. Therefore, the conveyance portions **54** and the barrier portions **55** can be exchanged and rearranged in accordance with, for example, the degree of kneading of raw materials, and the exchange and the rearrangement can be easily performed.

(93) Moreover, by constricting the tubes **39** in the axial direction of the second axial portion **41** and firmly affixing the end faces of the adjacent tubes **39** to each other, the path main body **64** of each of the paths **60** is formed, and the entrance **62** and the exit **63** of each of the paths **60** are integrally connected through the path main body **64**. Thus, in order to form the paths **60** in the screw main body **37**, it suffices that each of the tubes **39** having a length much shorter than the total length of the screw main body **37** is processed. Thus, the paths **60** are easily processed and handled when being formed.

(94) According to the continuous high shearing processing apparatus **1** having the above-described structure, the first extruder **2** preliminarily kneads resins. The resins melted by the kneading become raw materials having flowability, and are continuously supplied from the first extruder **2** to the conveyance path **53** through the supply port **34** of the second extruder **3**.

(95) As indicated by arrow B in FIG. **10**, the raw materials supplied to the second extruder **3** are introduced to the outer circumferential surface of the conveyance portion **54** positioned on the basal end side of the screw main body **37**. At this time, if the screw **21** rotates left-handed in an anticlockwise direction from the perspective of the basal end of the screw main body **37**, the flights **56** of the conveyance portions **54** continuously convey the raw materials toward the barrier portion **55** adjacent to the basal end of the screw main body **37** as indicated by solid-line arrows in FIG. **10**.

(96) At this time, shearing action, caused by a difference in speed between the flights **56** rotating along the conveyance path **53** and the inner circumferential surface of the cylinder portion **33**, is imparted to the raw materials, and the raw materials are stirred by a subtle twist of the flights **56**. As a result, the raw materials are kneaded thoroughly, and the dispersion of macromolecular components of the raw materials progresses.

(97) The raw materials, which have been subjected to shearing action, reach boundaries between the conveyance portions **54** and the barrier portions **55** along the conveyance path **53**. The flights **57** of the barrier portions **55** are twisted right-handed to convey raw materials from the basal end toward the tip of the screw main body **37** when the screw **21** rotates left-handed. As a result, the conveyance of raw materials is stopped by the flights **57**. In other words, when the screw **21** rotates left-handed, the flights **57** of the barrier portions **55** prevent raw materials from passing through the clearance between the barrier portions **55** and the inner circumferential surface of the cylinder portion **33** by limiting the flow of raw materials conveyed by the flights **56**.

(98) At this time, the pressure on the raw materials is increased at the boundaries between the conveyance portions **54** and the barrier portions **55**. Specifically, FIG. **11** shows the filling rate of raw materials at the places in the conveyance path **53**, which correspond to the conveyance portions **54** of the screw main body **37**, with gradations. That is, in the conveyance path **53**, the filling rate of raw materials becomes greater as the tone becomes darker. As is clear from FIG. **11**, the filling rate of raw materials becomes greater as they approach the barrier portions **55** in the conveyance path **53** corresponding to the conveyance portions **54**, and just before the barrier portions **55**, the filling rate of raw materials is 100%.

(99) Thus, just before the barrier portions **55**, a “raw-material receiver R” in which the filling rate of raw materials is 100% is formed. In the raw-material receiver R, the flow of raw materials is stopped, and thus, the pressure on the raw materials is increased. As indicated by broken-line arrows in FIG. **10** and FIG. **11**, the raw materials, the pressure on which has been increased, continuously flow into the path main body **64** from the entrance **62** opening in the outer circumferential surfaces of the conveyance portions **54**, and continuously flow through the path main body **64** from the basal end toward the tip of the screw main body **37**.

(100) As described above, the path sectional area defined by the bore of the path main body **64** is much smaller than the annular sectional area of the conveyance path **53** in the radial direction of the cylinder portion **33**. From another point of view, a widening area based on the bore of the path main body **64** is much smaller than that of the annular conveyance path **53**. Therefore, raw materials are rapidly squeezed when flowing from the entrance **62** into the path main body **64**, and

thus, extension action is imparted to the raw materials.

(101) Moreover, since the path sectional area is sufficiently smaller than the annular sectional area, raw materials collecting in the raw-material receiver R do not disappear. That is, some of the raw materials collecting in the raw-material receiver R continuously flow into the entrance **62**. In the meantime, new raw materials are fed toward the barrier portions **55** by the flights **56**. As a result, the filling rate just before the barrier portions **55** in the raw-material receiver R is thereby kept at 100% all the time. At this time, even if the amount of raw materials conveyed by the flights **56** somewhat changes, the change is absorbed by raw materials remaining in the raw-material receiver R. Raw materials can be thereby continuously and stably supplied to the paths **60**. Thus, in the paths **60**, extension action can be uninterruptedly and continuously imparted to the raw materials.

(102) The raw materials which have passed through the path main body **64** flow out of the exit **63** as indicated by solid-line arrows in FIG. **11**. The raw materials are thereby continuously returned to the other conveyance portions **54** adjacent to the barrier portions **55** in the direction of the tip of the screw main body **37**. The returned raw materials are continuously conveyed in the direction of the basal end of the screw main body **37** by the flights **56** of the conveyance portions **54**, and are subjected to shearing action again in the process of being conveyed. The raw materials, which have been subjected to shearing action, continuously flow into the path main body **64** from the entrance **62**, and are subjected to extension action again in the process of flowing through the path main body **64**.

(103) In the present embodiment, the conveyance portions **54** and the barrier portions **55** are alternately arranged in the axial direction of the screw main body **37**, and the paths **60** are arranged with a space therebetween in the axial direction of the screw main body **37**. Thus, the raw materials introduced to the screw main body **37** from the supply port **34** are continuously conveyed from the basal end toward the tip of the screw main body **37** while being alternately and repeatedly subjected to shearing action and extension action as indicated by arrows in FIG. **10** and FIG. **11**. Thus, the degree of kneading of raw materials is increased, and the dispersion of macromolecular components of the raw materials is promoted.

(104) The raw materials which have reached the tip of the screw main body **37** become sufficiently kneaded materials, and flow out of the exit **63** of each of the paths **60**. The kneaded materials which have flowed out are continuously conveyed to a gap between the cylinder portion **33** and the head portion **36** by the flight **58** of the discharge conveyance portion **59**, and then continuously supplied to the third extruder **4** from the discharge port **36a**.

(105) As already described, in the third extruder **4**, gaseous materials and other volatile components included in kneaded materials are continuously removed from the kneaded materials. The kneaded materials, from which gaseous materials and other volatile components are removed, are continuously discharged from the discharge port **28** of the head portion **27** to the outside of the high shearing processing apparatus **1**. The discharged kneaded materials are soaked in cooling water stored in a water tank. The kneaded materials are thereby forcibly cooled, and a desired resin molding is obtained.

(106) As described above, according to the first embodiment, in the second extruder **3**, raw materials supplied from the first extruder **2** are conveyed while being reversed more than once in the axial direction of the screw main body **37**, and in the process of conveyance, shearing action and extension action are repeatedly imparted to the raw materials. In other words, raw materials do not circulate through the same places on the outer circumferential surface of the screw main body **37** many times, and thus can be uninterruptedly supplied to the third extruder **4** from the second extruder **3**.

(107) Accordingly, kneaded materials that are sufficiently kneaded can be continuously molded, and the production efficiency of kneaded materials can be dramatically increased as compared to that of a batch extruder.

(108) Along with this, in the present embodiment, resin preliminarily kneaded by the first extruder

2 continues being uninterruptedly supplied to the second extruder 3. Thus, the flow of resin does not temporarily stagnate inside the first extruder 2. Temperature change, viscosity change, or phase change of the resin, caused when the kneaded resin stagnates inside the first extruder 2, can be thereby prevented. As a result, raw materials having uniform quality all the time can be supplied to the second extruder 3 from the first extruder 2.

(109) Moreover, according to the first embodiment, the completely continuous production, not apparently continuous production, of kneaded materials is enabled. That is, shearing action and extension action can be alternately imparted to raw materials in the second extruder 3 while raw materials are uninterruptedly and continuously conveyed from the first extruder 2 to the second extruder 3 and the third extruder 4. According to the above-described structure, raw materials in a melted state can be stably supplied from the first extruder 2 to the second extruder 3.

(110) Moreover, according to the first embodiment, in the completely continuous production, the optimum operating conditions can be set for the first extruder 2 and the second extruder 3, while their operating conditions are associated with each other. For example, if resin is preliminarily kneaded with the first extruder 2, the screw rotational rate can be set at 100 to 300 rpm as it has been conventionally set. Thus, the resin can be sufficiently heated and melted, and preliminarily kneaded. On the other hand, in the second extruder 3, the screw 21 can be rotated at a high rate of 600 to 3,000 rpm. Thus, shearing action and extension action can be alternately and effectively imparted to the resin.

(111) Accordingly, it suffices that the first extruder 2 and the second extruder 3 comprise screws according to the respective roles or functions. That is, it suffices that the first extruder 2 comprises the screws 7a and 7b according to the role or function of preliminarily kneading supplied materials. On the other hand, it suffices that the second extruder 3 comprises the screw 21 according to the role or function of imparting shearing action and extension action to raw materials in a melted state supplied from the first extruder 2. The first extruder 2 and the second extruder 3 can be thereby prevented from being elongated.

(112) Furthermore, in the screw 21, the conveyance portions 54, the barrier portions 55, and the paths 60 are disposed in combination without a plasticization zone provided in a screw of a conventional single screw extruder. Thus, the second extruder 3 can be easily operated.

(113) In addition, the root diameter of the screw 21 is set at a fixed value over the total length of the screw 21. Thus, the conveyance path 53 for conveying raw materials has uniform annular sectional shapes over the total length of the screw 21. When shearing action and extension action are alternately imparted to raw materials, they can be sequentially and smoothly imparted, and uniform kneading can be performed.

(114) Here, results of a high dispersion verification test performed on kneaded materials in the case where raw materials are kneaded while shearing action and extension action are alternately imparted to them by the above-described completely continuous production will be described.

(115) In the test, two kinds of materials, a polycarbonate (PC) resin and a polymethyl methacrylate (PMMA) resin, are supplied to the first extruder 2, in which the effective length (L/D) of the kneading portion 12 with respect to the screw effective length (L/D) 50 is set at 7.9, and materials in a melted state are produced by preliminarily kneading them. In addition, the materials in a melted state are continuously supplied from the first extruder 2 to the second extruder 3 as raw materials of the second extruder 3.

(116) In the test, the screw 21 is configured to repeat the above-described shearing and extension operation ten times. Further, the specifications of the screw 21 are set as follows: the screw diameter is set at 36 mm, the screw effective length (L/D) is set at 25, the screw rotational rate is set at 1,400 rpm, the supply of raw materials is set at 1.4 kg/h, and the barrel set temperature is set at 260° C.

(117) Through the above-described test, intended transparent kneaded materials were continuously obtained.

(118) According to the first embodiment, the paths **60** imparting extension action to raw materials extend in the axial direction of the screw main body **37** at positions eccentric to the axial line **O1**, which is the center of rotation of the screw main body **37**. Thus, the paths **60** revolve around the axial line **O1**. In other words, the tubular wall surfaces **61** defining the paths **60** revolve around the axial line **O1** without rotating on the axial line **O1**.

(119) Accordingly, when raw materials pass through the paths **60**, the raw materials are not actively stirred inside the paths **60** although they are subjected to centrifugal force. Therefore, the raw materials passing through the paths **60** are hardly subjected to shearing action, and the raw materials passing through the paths **60** and returning to the outer circumferential surfaces of the conveyance portions **54** are mainly subjected to extension action.

(120) Therefore, according to the screw **21** of the first embodiment, places where shearing action is imparted to raw materials and places where extension action is imparted to raw materials can be clearly determined. For this reason, a structure advantageous in ascertaining the degree of kneading of raw materials is achieved, and the degree of kneading can be accurately controlled. As a result, kneaded materials having a microscopic dispersion structure in which macromolecular components of raw materials are dispersed at a nanolevel can be produced.

(121) In addition, since all the paths **60** are eccentric to the axial line **O1**, extension action can be equally imparted to raw materials passing through the paths **60**. That is, the discrepancy in kneading conditions between the paths **60** can be resolved, and uniform kneading can be performed.

Second Embodiment

(122) FIG. **12** shows a second embodiment. The second embodiment differs from the first embodiment in the matters related to an axis of rotation **38**. The other structures of a screw **21** are basically the same as those of the first embodiment. Therefore, in the second embodiment, the same structural portions as those of the first embodiment will be given the same reference numbers, and description thereof will be omitted.

(123) As shown in FIG. **12**, a refrigerant path **71** is formed inside the axis of rotation **38**. The refrigerant path **71** coaxially extends along an axial line **O1** of the axis of rotation **38**. One end of the refrigerant path **71** is connected to an exit pipe **73** through a rotary joint **72** at a joint portion **42**. The other end of the refrigerant path **71** is liquid-tightly closed by the tip of the axis of rotation **38**.

(124) A refrigerant introduction pipe **74** is coaxially inserted in the refrigerant path **71**. One end of the refrigerant introduction pipe **74** is connected to an entrance pipe **75** through the rotary joint **72**. The other end of the refrigerant introduction pipe **74** is opened in the refrigerant path **71** near the other end of the refrigerant path **71**.

(125) In the second embodiment, a refrigerant such as water or oil is fed from the entrance pipe **75** to the refrigerant path **71** through the rotary joint **72** and the refrigerant introduction pipe **74**. The refrigerant fed into the refrigerant path **71** returns to the joint portion **42** of the axis of rotation **38** through a gap between the inner circumferential surface of the refrigerant path **71** and the outer circumferential surface of the refrigerant introduction pipe **74**, and is returned to the exit pipe **73** through the rotary joint **72**.

(126) According to the second embodiment, the refrigerant circulates in the axial direction of the axis of rotation **38**, and thus, a screw main body **37** can be refrigerated by the refrigerant. Therefore, the temperature of the screw main body **37** in contact with raw materials can be properly adjusted, and deterioration, change in viscosity, etc., of resin due to a rise in the temperature of the raw materials can be prevented.

Third Embodiment

(127) FIG. **13** to FIG. **18** show a third embodiment. The third embodiment differs from the first embodiment in the matters related to a screw main body **37**. The other structures of a screw **21** are basically the same as those of the first embodiment. Therefore, in the third embodiment, the same structural portions as those of the first embodiment will be given the same reference numbers, and

description thereof will be omitted.

(128) As shown in FIG. 13 and FIG. 14, cylindrical tubes 39 constituting the screw main body 37 are constricted in the axial direction of a second axial portion 41 between a first collar 44 and a second collar 51, and the end faces of the adjacent tubes 39 are firmly affixed to each other without any gap as in the first embodiment.

(129) At this time, all the tubes 39 are coaxially joined on the second axial portion 41, and each of the tubes 39 and an axis of rotation 38 are integrally assembled. This makes it possible to rotate each of the tubes 39 on an axial line O1 together with the axis of rotation 38, that is, rotate the screw main body 37 on the axial line O1.

(130) In such a state, each of the tubes 39 serves as a structural element that defines the outside diameter D1 (see FIG. 15) of the screw main body 37. That is, the outside diameters D1 of the tubes 39 coaxially joined along the second axial portion 41 are set to be equal to each other. The outside diameter D1 of the screw main body 37 (each of the tubes 39) is defined as a diameter passing through the axial line O1, which is the center of rotation of the axis of rotation 38.

(131) The segmental screw 21 in which the outside diameter D1 of the screw main body 37 (each of the tubes 39) is a fixed value is thereby formed. In the segmental screw 21, screw elements can be held in free order and combination along the axis of rotation 38 (that is, the second axial portion 41). With respect to the screw elements, for example, each of the tubes 39 on which at least parts of flights 84 and 86, which will be described later, are formed can be defined as one screw element.

(132) In this manner, by segmenting the screw 21, its convenience can be significantly improved with respect to, for example, changes and adjustments to the specifications or the upkeep and maintenance of the screw 21.

(133) Moreover, the segmental screw 21 is coaxially accommodated in a cylinder portion 33 of a barrel 20. Specifically, the screw main body 37 with the screw elements held along the axis of rotation 38 (the second axial portion 41) is rotatably accommodated in the cylinder portion 33. In this state, a first axial portion 40 (a joint portion 42 and a stopper portion 43) of the axis of rotation 38 projects from one end portion of the barrel 20 to the outside of the barrel 20.

(134) Moreover, in this state, a conveyance path 53 for conveying raw materials is formed between the outer circumferential surface in the circumferential direction of the screw main body 37 and the inner circumferential surface of the cylinder portion 33. The conveyance path 53 has an annular sectional shape in the radial direction of the cylinder portion 33, and extends in the axial direction of the cylinder portion 33.

(135) As shown in FIG. 13 to FIG. 16, the screw main body 37 comprises conveyance portions 81 for conveying raw materials and barrier portions 82 for limiting the flow of raw materials. To be specific, conveyance portions 81 are disposed at the basal end of the screw main body 37, which corresponds to the one end portion of the barrel 20, and conveyance portions 81 are disposed at the tip of the screw main body 37, which corresponds to the other end portion of the barrel 20.

Moreover, between the conveyance portions 81, conveyance portions 81 and the barrier portions 82 are alternately disposed in the axial direction from the basal end toward the tip of the screw main body 37.

(136) A supply port 34 of the barrel 20 opens toward the conveyance portion 81 disposed on the basal end side of the screw main body 37.

(137) Each of the conveyance portions 81 comprises the flight 84 spirally twisted. The flight 84 projects from the outer circumferential surface in the circumferential direction of the tubes 39 toward the conveyance path 53. The flight 84 is twisted to convey raw materials from the basal end toward the tip of the screw main body 37 when the screw 21 rotates left-handed in an anticlockwise direction from the perspective of the basal end of the screw main body 37. That is, the flight 84 is twisted right-handed as in the case of a right-handed screw.

(138) Each of the barrier portions 82 comprises the flight 86 spirally twisted. The flight 86 projects from the outer circumferential surface in the circumferential direction of the tubes 39 toward the

conveyance path **53**. The flight **86** is twisted to convey raw materials from the tip toward the basal end of the screw main body **37** when the screw **21** rotates left-handed in an anticlockwise direction from the perspective of the basal end of the screw main body **37**. That is, the flight **86** is twisted left-handed as in the case of a left-handed screw.

(139) The twist pitch of the flight **86** of each of the barrier portions **82** is set to be smaller than or equal to that of the flight **84** of each of the conveyance portions **81**. Moreover, a slight clearance is secured between the apexes of the flights **84** and **86** and the inner circumferential surface of the cylinder portion **33** of the barrel **20**. In this case, the clearance between the outside diameter portions of the barrier portions **82** (the apexes of the flights **86**) and the inner circumferential surface of the cylinder portion **33** is preferably set to be within a range of 0.1 to 2 mm. More preferably, the clearance is set to be within a range of 0.1 to 0.7 mm. This can surely limit the conveyance of raw materials through the clearance.

(140) Here, the lengths of the conveyance portions **81** in the axial direction of the screw main body **37** are set as appropriate in accordance with, for example, the kind of raw materials, the degree of kneading of raw materials, and the production of kneaded materials per unit time. The conveyance portions **81** are at least areas where the flights **84** are formed on the outer circumferential surfaces of the tubes **39**, but are not limited to areas between the start points and the end points of the flights **84**.

(141) That is, areas outside the flights **84** of the outer circumferential surfaces of the tubes **39** may be regarded as the conveyance portions **81**. For example, if a cylindrical spacer or a cylindrical collar is disposed at a position adjacent to the tubes **39** comprising the flights **84**, the spacer or the collar also can be included in the conveyance portions **81**.

(142) Moreover, the lengths of the barrier portions **82** in the axial direction of the screw main body **37** are set as appropriate in accordance with, for example, the kind of raw materials, the degree of kneading of raw materials, and the production of kneaded materials per unit time. The barrier portions **82** function to stop the flow of raw materials fed by the conveyance portions **81**. That is, the barrier portions **82** are adjacent to the conveyance portions **81** on the downstream side in the conveyance direction of raw materials, and configured to prevent raw materials fed by the conveyance portions **81** from passing through the clearance between the apexes of the flights **86** and the inner circumferential surface of the cylinder portion **33**.

(143) Moreover, in the above-described screw **21**, each of the flights **84** and **86** projects from the outer circumferential surfaces of the tubes **39** having the outside diameters **D1** equal to each other toward the conveyance path **53**. Thus, the outer circumferential surface in the circumferential direction of each of the tubes **39** defines the root diameter of the screw **21**. The root diameter of the screw **21** is kept at a fixed value over the total length of the screw **21**.

(144) As shown in FIG. **13**, FIG. **14**, and FIG. **17**, the screw main body **37** comprises paths **88** extending in the axial direction of the screw main body **37**. If one barrier portion **82** and two conveyance portions **81**, between which the barrier portion **82** is sandwiched, are regarded as one unit, a path **88** is formed over tubes **39** of both of the conveyance portions **81** and a tube **39** of the barrier portion **82**. In this case, the paths **88** are arranged at predetermined intervals (for example, regular intervals) in the same straight line in the axial direction of the screw main body **37**.

(145) Moreover, the paths **88** are provided at positions eccentric to the axial line **O1** of the axis of rotation **38** inside the tubes **39**. In other words, the paths **88** are shifted from the axial line **O1**, and revolve around the axial line **O1** when the screw main body **37** rotates.

(146) As shown in FIG. **15**, the paths **88** are, for example, holes having a circular sectional shape. The inside diameter of the holes is, for example, set to be greater than or equal to 1 mm but less than 6 mm, and preferably, greater than or equal to 1 mm but less than 5 mm. Moreover, the tubes **39** of the conveyance portions **81** and the barrier portions **82** comprise tubular wall surfaces **89** defining the holes. That is, the paths **88** are holes composed of hollow spaces only, and the wall surfaces **89** continuously surround the hollow paths **88** in the circumferential direction. The paths

88 are thereby formed as hollow spaces which allow only the flow of raw materials. In other words, inside the paths **88**, there are no other elements constituting the screw main body **37**. Moreover, the wall surfaces **89** revolve around the axial line **O1** without rotating on the axial line **O1**, when the screw main body **37** rotates.

(147) As shown in FIG. 13, FIG. 14, and FIG. 18, each of the paths **88** comprises an entrance **91**, an exit **92**, and a path main body **93** connecting the entrance **91** and the exit **92**. The entrance **91** and the exit **92** are provided close to both sides of one barrier portion **82**. From another point of view, in one conveyance portion **81** adjacent to two barrier portions **82** adjacent to each other, the entrance **91** is opened in the outer circumferential surface near the downstream end of the conveyance portion **81**, and the exit **92** is opened in the outer circumferential surface near the upstream end of the conveyance portion **81**.

(148) The path main body **93** extends straight without branching on the way in the axial direction of the screw main body **37**. The figures show, as an example, a state in which the path main body **93** extends parallel to the axial line **O1**. Both sides of the path main body **93** are closed in the axial direction.

(149) The entrance **91** is provided on one side of the path main body **93**, that is, a portion closer to the basal end of the screw main body **37**. In this case, the entrance **91** may be opened in the outer circumferential surface of the screw main body **37** from an end face on the one side of the path main body **93**, or may be opened in the outer circumferential surface of the screw main body **37** from a portion closer to the end face on the one side of the path main body **93**, that is, a portion located short of the end face. The opening direction of the entrance **91** is not limited to those orthogonal to the axial line **O1**, but may be those crossing the axial line **O1**. In this case, entrances **91** may be provided by opening one side of the path main body **93** in directions.

(150) The exit **92** is provided on the other side (the opposite side to the one side) of the path main body **93**, that is, a portion closer to the tip of the screw main body **37**. In this case, the exit **92** may be opened in the outer circumferential surface of the screw main body **37** from an end face on the other side of the path main body **93**, or may be opened in the outer circumferential surface of the screw main body **37** from a portion closer to the end face on the other side of the path main body **93**, that is, a portion located short of the end face. The opening direction of the exit **92** is not limited to those orthogonal to the axial line **O1**, but may be those crossing the axial line **O1**. In this case, exits **92** may be provided by opening one side of the path main body **93** in directions.

(151) The path main body **93** connecting the entrance **91** and the exit **92** traverses the barrier portion **82** of the above-described one unit, and has a length stretching between the two conveyance portions **81**, between which the barrier portion **82** is sandwiched. In this case, the bore of the path main body **93** may be set to be smaller than those of the entrance **91** and the exit **92**, or may be set to be equal to them. In either case, the path sectional area defined by the bore of the path main body **93** is set to be much smaller than the annular sectional area in the radial direction of the above-described annular conveyance path **53**.

(152) In the present embodiment, if the screw **21** is disassembled by detaching the tubes **39** on which the flights **84** and **86** are formed from the axis of rotation **38**, the tubes **39** on which at least parts of the flights **84** and **86** are formed can also be referred to as the above-described screw elements.

(153) Thus, the screw main body **37** of the screw **21** can be formed by sequentially disposing the tubes **39** as the screw elements on the outer circumference of the axis of rotation **38**. Therefore, the conveyance portions **81** and the barrier portions **82** can be exchanged and rearranged in accordance with, for example, the degree of kneading of raw materials, and the exchange and the rearrangement can be easily performed.

(154) Moreover, by constricting the tubes **39** in the axial direction of the second axial portion **41** and firmly affixing the end faces of the tubes **39** to each other, the path main body **93** of each of the paths **88** is formed, and the entrance **91** and the exit **92** of each of the paths **88** are integrally

connected through the path main body **93**. Thus, in order to form the paths **88** in the screw main body **37**, it suffices that each of the tubes **39** having a length much shorter than the total length of the screw main body **37** is processed. Thus, the paths **88** are easily processed and handled when being formed.

(155) According to a continuous high shearing processing apparatus **1** having the above-described structure, a first extruder **2** preliminarily kneads resins. The resins melted by the kneading become raw materials having flowability, and are continuously supplied from the first extruder **2** to the conveyance path **53** through the supply port **34** of a second extruder **3**.

(156) As indicated by arrow C in FIG. **17**, the raw materials supplied to the second extruder **3** are introduced to the outer circumferential surface of the conveyance portion **81** positioned on the basal end side of the screw main body **37**. At this time, if the screw **21** rotates left-handed in an anticlockwise direction from the perspective of the basal end of the screw main body **37**, the flights **84** of the conveyance portions **81** continuously convey the raw materials toward the tip of the screw main body **37** as indicated by solid-line arrows in FIG. **17**.

(157) At this time, shearing action, caused by a difference in speed between the flights **84** rotating along the conveyance path **53** and the inner circumferential surface of the cylinder portion **33**, is imparted to the raw materials, and the raw materials are stirred by a subtle twist of the flights **84**. As a result, the raw materials are kneaded thoroughly, and the dispersion of macromolecular components of the raw materials progresses.

(158) The raw materials, which have been subjected to shearing action, reach boundaries between the conveyance portions **81** and the barrier portions **82** along the conveyance path **53**. The flights **86** of the barrier portions **82** are twisted left-handed to convey raw materials from the tip toward the basal end of the screw main body **37** when the screw **21** rotates left-handed. As a result, the conveyance of raw materials is stopped by the flights **86**. In other words, when the screw **21** rotates left-handed, the flights **86** of the barrier portions **82** prevent raw materials from passing through the clearance between the barrier portions **82** and the inner circumferential surface of the cylinder portion **33** by limiting the flow of raw materials conveyed by the flights **84**.

(159) At this time, the pressure on the raw materials is increased at the boundaries between the conveyance portions **81** and the barrier portions **82**. Specifically, FIG. **18** shows the filling rate of raw materials at the places in the conveyance path **53**, which correspond to the conveyance portions **81** of the screw main body **37**, with gradations. That is, in the conveyance path **53**, the filling rate of raw materials becomes greater as the tone becomes darker. As is clear from FIG. **18**, the filling rate of raw materials becomes greater as they approach the barrier portions **82** in the conveyance path **53** corresponding to the conveyance portions **81**, and just before the barrier portions **82**, the filling rate of raw materials is 100%.

(160) Thus, just before the barrier portions **82**, a “raw-material receiver R” in which the filling rate of raw materials is 100% is formed. In the raw-material receiver R, the flow of raw materials is stopped, and thus, the pressure on the raw materials is increased. As indicated by broken-line arrows in FIG. **17** and FIG. **18**, the raw materials, the pressure on which has been increased, continuously flow into the path main body **93** from the entrance **91** opening in the downstream ends of the conveyance portions **81**, and continuously flow through the path main body **93** from the basal end toward the tip of the screw main body **37**.

(161) As described above, the path sectional area defined by the bore of the path main body **93** is much smaller than the annular sectional area of the conveyance path **53** in the radial direction of the cylinder portion **33**. From another point of view, a widening area based on the bore of the path main body **93** is much smaller than that of the annular conveyance path **53**. Therefore, raw materials are rapidly squeezed when flowing from the entrance **91** into the path main body **93**, and thus, extension action is imparted to the raw materials.

(162) Moreover, since the path sectional area is sufficiently smaller than the annular sectional area, raw materials collecting in the raw-material receiver R do not disappear. That is, some of the raw

materials collecting in the raw-material receiver R continuously flow into the entrance **91**. In the meantime, new raw materials are fed toward the barrier portions **82** by the flights **84**. As a result, the filling rate just before the barrier portions **82** in the raw-material receiver R is thereby kept at 100% all the time. At this time, even if the amount of raw materials conveyed by the flights **84** somewhat changes, the change is absorbed by raw materials remaining in the raw-material receiver R. Raw materials can be thereby continuously and stably supplied to the paths **88**. Thus, in the paths **88**, extension action can be uninterruptedly and continuously imparted to the raw materials. (163) The raw materials which have passed through the path main body **93** flow out of the exit **92** as indicated by solid-line arrows in FIG. **18**. The raw materials are thereby continuously returned to the outer circumferential surfaces of the other conveyance portions **81** adjacent to the barrier portions **82** on the tip side of the screw main body **37**. The returned raw materials are continuously conveyed in the direction of the tip of the screw main body **37** by the flights **84** of the other conveyance portions **81**, and are subjected to shearing action again in the process of being conveyed. The raw materials, which have been subjected to shearing action, continuously flow into the path main body **93** from the entrance **91**, and are subjected to extension action again in the process of flowing through the path main body **93**.

(164) In the present embodiment, the conveyance portions **81** and the barrier portions **82** are alternately arranged in the axial direction of the screw main body **37**, and the paths **88** are arranged with a space therebetween in the axial direction of the screw main body **37**. Thus, the raw materials introduced to the screw main body **37** from the supply port **34** are continuously conveyed from the basal end toward the tip of the screw main body **37** while being alternately and repeatedly subjected to shearing action and extension action as indicated by arrows in FIG. **17** and FIG. **18**. Thus, the degree of kneading of raw materials is increased, and the dispersion of macromolecular components of the raw materials is promoted.

(165) Then, the raw materials which have reached the tip of the screw main body **37** become sufficiently kneaded materials, and are continuously supplied to a third extruder **4** from a discharge port **36a**, and gaseous materials and other volatile components included in the kneaded materials are continuously removed from the kneaded materials.

(166) As described above, according to the third embodiment, the completely continuous production, not apparently continuous production, of kneaded materials is enabled. That is, resin preliminarily kneaded by the first extruder **2** continues being uninterruptedly supplied to the second extruder **3**, and thus, the flow of resin does not temporarily stagnate inside the first extruder **2**. Thus, temperature change, viscosity change, or phase change of the resin, caused when the kneaded resin stagnates inside the first extruder **2**, can be prevented. As a result, raw materials having uniform quality all the time can be supplied to the second extruder **3** from the first extruder **2**.

(167) Furthermore, according to the third embodiment, the lengths of shearing action regions and extension action regions for raw materials in the axial direction can be set individually. Thus, the optimum number of times shearing action and extension action are imparted and the optimum time for which shearing action and extension action are imparted for kneading raw materials can be set.

(168) Moreover, according to the third embodiment, the paths **88** imparting extension action to raw materials extend in the axial direction of the screw main body **37** at positions eccentric to the axial line O**1**, which is the center of rotation of the screw main body **37**. Thus, the paths **88** revolve around the axial line O**1**. In other words, the tubular wall surfaces **89** defining the paths **88** revolve around the axial line O**1** without rotating on the axial line O**1**.

(169) Accordingly, when raw materials pass through the paths **88**, the raw materials are not actively stirred inside the paths **88**. Therefore, the raw materials passing through the paths **88** are hardly subjected to shearing action, and the raw materials passing through the paths **88** and returning to the outer circumferential surfaces of the conveyance portions **81** are mainly subjected to extension action. Therefore, also in the screw **21** of the third embodiment, places where shearing action is imparted to raw materials and places where extension action is imparted to raw materials can be

clearly determined.

(170) Here, results of a high dispersion verification test performed on kneaded materials in the case where raw materials are kneaded while shearing action and extension action are alternately imparted to them by the above-described completely continuous production will be described.

(171) In the test, two kinds of materials, a polycarbonate (PC) resin and a polymethyl methacrylate (PMMA) resin, are supplied to the first extruder **2**, in which the effective length (L/D) of a kneading portion **12** with respect to the screw effective length (L/D) **50** is set at 7.9, and materials in a melted state are produced by preliminarily kneading them. In addition, the materials in a melted state are continuously supplied from the first extruder **2** to the second extruder **3** as raw materials of the second extruder **3**.

(172) In the test, the screw **21** is configured to repeat the above-described shearing and extension operation eight times. Further, the specifications of the screw **21** are set as follows: the screw diameter is set at 36 mm, the screw effective length (L/D) is set at 16.7, the screw rotational rate is set at 2,300 rpm, the supply of raw materials is set at 10.0 kg/h, and the barrel set temperature is set at 240° C.

(173) Through the above-described test, intended transparent kneaded materials were continuously obtained.

Fourth Embodiment

(174) FIG. **19** to FIG. **27** show a fourth embodiment. The fourth embodiment differs from the first embodiment in the matters related to a screw main body **37**. The other structures of a screw **21** are basically the same as those of the first embodiment. Therefore, in the fourth embodiment, the same structural portions as those of the first embodiment will be given the same reference numbers, and description thereof will be omitted.

(175) As shown in FIG. **19** to FIG. **21**, cylindrical tubes **39** constituting the screw main body **37** are constricted in the axial direction of a second axial portion **41** between a first collar **44** and a second collar **51**, and the end faces of the adjacent tubes **39** are firmly affixed to each other without any gap as in the first embodiment.

(176) At this time, all the tubes **39** are coaxially joined on the second axial portion **41**, and each of the tubes **39** and an axis of rotation **38** are integrally assembled. This makes it possible to rotate each of the tubes **39** on an axial line **O1** together with the axis of rotation **38**, that is, rotate the screw main body **37** on the axial line **O1**.

(177) In such a state, each of the tubes **39** serves as a structural element that defines the outside diameter **D1** (see FIG. **22**) of the screw main body **37**. That is, the outside diameters **D1** of the tubes **39** coaxially joined along the second axial portion **41** are set to be equal to each other. The outside diameter **D1** of the screw main body **37** (each of the tubes **39**) is defined as a diameter passing through the axial line **O1**, which is the center of rotation of the axis of rotation **38**.

(178) The segmental screw **21** in which the outside diameter **D1** of the screw main body **37** (each of the tubes **39**) is a fixed value is thereby formed. In the segmental screw **21**, screw elements can be held in free order and combination along the axis of rotation **38** (that is, the second axial portion **41**). With respect to the screw elements, for example, each of the tubes **39** on which at least parts of flights **105**, **107**, **110**, **111**, and **112**, which will be described later, are formed can be defined as one screw element.

(179) In this manner, by segmenting the screw **21**, its convenience can be significantly improved with respect to, for example, changes and adjustments to the specifications or the upkeep and maintenance of the screw **21**.

(180) Moreover, the segmental screw **21** is coaxially accommodated in a cylinder portion **33** of a barrel **20**. Specifically, the screw main body **37** with the screw elements held along the axis of rotation **38** (the second axial portion **41**) is rotatably accommodated in the cylinder portion **33**. In this state, a first axial portion **40** (a joint portion **42** and a stopper portion **43**) of the axis of rotation **38** projects from one end portion of the barrel **20** to the outside of the barrel **20**.

(181) Moreover, in this state, a conveyance path **53** for conveying raw materials is formed between the outer circumferential surface in the circumferential direction of the screw main body **37** and the inner circumferential surface of the cylinder portion **33**. The conveyance path **53** has an annular sectional shape in the radial direction of the cylinder portion **33**, and extends in the axial direction of the cylinder portion **33**.

(182) As shown in FIG. **19** to FIG. **21**, the screw main body **37** comprises conveyance portions **101** for conveying raw materials, barrier portions **102** for limiting the flow of raw materials, and circulation portions **103** temporarily circulating raw materials. To be specific, conveyance portions **101** are disposed at the basal end of the screw main body **37**, which corresponds to the one end portion of the barrel **20**, and conveyance portions **101** are disposed at the tip of the screw main body **37**, which corresponds to the other end portion of the barrel **20**. Moreover, between the conveyance portions **101**, the circulation portions **103** and the barrier portions **102** are alternately disposed in the axial direction from the basal end toward the tip of the screw main body **37**.

(183) A supply port **34** of the barrel **20** opens toward the conveyance portion **101** disposed on the basal end side of the screw main body **37**.

(184) Each of the conveyance portions **101** comprises the flight **105** spirally twisted. The flight **105** projects from the outer circumferential surface in the circumferential direction of the tubes **39** toward the conveyance path **53**. The flight **105** is twisted to convey raw materials from the basal end toward the tip of the screw main body **37** when the screw **21** rotates left-handed in an anticlockwise direction from the perspective of the basal end of the screw main body **37**. That is, the flight **105** is twisted right-handed as in the case of a right-handed screw.

(185) Each of the barrier portions **102** comprises the flight **107** spirally twisted. The flight **107** projects from the outer circumferential surface in the circumferential direction of the tubes **39** toward the conveyance path **53**. The flight **107** is twisted to convey raw materials from the tip toward the basal end of the screw main body **37** when the screw **21** rotates left-handed in an anticlockwise direction from the perspective of the basal end of the screw main body **37**. That is, the flight **107** is twisted left-handed as in the case of a left-handed screw.

(186) The circulation portions **103** are adjacent to the barrier portions **102** from the basal end side of the axis of rotation **38**. Each of the circulation portions **103** comprises the first to third flights **110**, **111**, and **112** spirally twisted. The first to third flights **110**, **111**, and **112** each project from the outer circumferential surface in the circumferential direction of the tubes **39** toward the conveyance path **53**.

(187) The first to third flights **110**, **111**, and **112** are disposed to be adjacent to each other in the axial direction of the screw main body **37**. The first to third flights **110**, **111**, and **112** are twisted to convey raw materials from the basal end toward the tip of the screw main body **37** when the screw **21** rotates left-handed in an anticlockwise direction from the perspective of the basal end of the screw main body **37**. That is, the first to third flights **110**, **111**, and **112** are twisted right-handed as in the case of a right-handed screw.

(188) In this case, the twist pitch of the flight **107** of each of the barrier portions **102** is set to be smaller than or equal to those of the flights **105** of the conveyance portions **101** and the flights **110**, **111**, and **112** of the circulation portions **103**. Furthermore, the twist pitch of the second flights **111** is set to be smaller than those of the first and third flights **110** and **112**. Moreover, a slight clearance is secured between the apexes of the flights **105**, **107**, **110**, **111**, and **112** and the inner circumferential surface of the cylinder portion **33** of the barrel **20**.

(189) Further, of the first to third flights **110**, **111**, and **112**, the third flights **112** are disposed on the upstream side in the conveyance direction, and the first flights **110** are disposed on the downstream side in the conveyance direction. The second flights **111** are disposed between the third flights **112** and the first flights **110**.

(190) In the present embodiment, each of the barrier portions **102** is designed, such that raw materials can flow over each of the barrier portions **102**. Specifically, each of the barrier portions

102 is designed, such that raw materials can pass through a space between each of the barrier portions **102** and the cylinder portion **33** in a state in which the screw **21** is rotatably inserted in the cylinder portion **33** of the barrel **20**. In this case, the clearance between the outside diameter portion of each of the barrier portions **102** (the apexes of the flights **107**) and the inner circumferential surface of the cylinder portion **33** is preferably set to be within a range of 0.1 to 3 mm. More preferably, the clearance is set to be within a range of 0.1 to 1.5 mm.

(191) Here, the lengths of the conveyance portions **101** in the axial direction of the screw main body **37** are set as appropriate in accordance with, for example, the kind of raw materials, the degree of kneading of raw materials, and the production of kneaded materials per unit time. The conveyance portions **101** are at least areas where the flights **105** are formed on the outer circumferential surfaces of the tubes **39**, but are not limited to areas between the start points and the end points of the flights **105**.

(192) That is, areas outside the flights **105** of the outer circumferential surfaces of the tubes **39** may be regarded as the conveyance portions **101**. For example, if a cylindrical spacer or a cylindrical collar is disposed at a position adjacent to the tubes **39** comprising the flights **101**, the spacer or the collar also can be included in the conveyance portions **101**.

(193) Moreover, the lengths of the barrier portions **102** in the axial direction of the screw main body **37** are set as appropriate in accordance with, for example, the kind of raw materials, the degree of kneading of raw materials, and the production of kneaded materials per unit time. The barrier portions **102** according to the present embodiment function to stop the flow of raw materials fed by the conveyance portions **101** and allow some of the raw materials to flow over the barrier portions **102**.

(194) Moreover, in the above-described screw **21**, each of the flights **105**, **107**, **110**, **111**, and **112** projects from the outer circumferential surfaces of the tubes **39** having the outside diameters **D1** equal to each other toward the conveyance path **53**. Thus, the outer circumferential surface in the circumferential direction of each of the tubes **39** defines the root diameter of the screw **21**. The root diameter of the screw **21** is kept at a fixed value over the total length of the screw **21**.

(195) As shown in FIG. **19** to FIG. **21**, the screw main body **37** comprises paths **115** extending in the axial direction of the screw main body **37**. The paths **115** are formed in the tubes **39** of the respective circulation portions **103**. In this case, the paths **115** are arranged at predetermined intervals (for example, regular intervals) in the same straight line in the axial direction of the screw main body **37**.

(196) Moreover, the paths **115** are provided at positions eccentric to the axial line **O1** of the axis of rotation **38** inside the tubes **39**. In other words, the paths **115** are shifted from the axial line **O1**, and revolve around the axial line **O1** when the screw main body **37** rotates.

(197) As shown in FIG. **22**, the paths **115** are, for example, holes having a circular sectional shape. The inside diameter of the holes is, for example, set to be greater than or equal to 1 mm but less than 6 mm, and preferably, greater than or equal to 1 mm but less than 5 mm. Moreover, the tubes **39** of the circulation portions **103** comprise tubular wall surfaces **116** defining the holes. That is, the paths **115** are holes composed of hollow spaces only, and the wall surfaces **116** continuously surround the hollow paths **115** in the circumferential direction. The paths **115** are thereby formed as hollow spaces which allow only the flow of raw materials. In other words, inside the paths **115**, there are no other elements constituting the screw main body **37**. Moreover, the wall surfaces **116** revolve around the axial line **O1** without rotating on the axial line **O1**, when the screw main body **37** rotates.

(198) As shown in FIG. **19**, FIG. **20**, and FIG. **27**, each of the paths **115** comprises an entrance **117**, an exit **118**, and a path main body **119** connecting the entrance **117** and the exit **118**. The entrance **117** and the exit **118** are opened in the outer circumferential surfaces of the tubes **39** constituting the circulation portions **103**. The figures show an example of the paths **115**. In the paths **115**, the path main body **119** is provided in the tubes **39** on which the first flights **110** are formed, and the

entrance **117** and the exit **118** are opened in the outer circumferential surfaces of the tubes **39**. The positions where the entrance **117** and the exit **118** are opened can be freely set within the outer circumferential surfaces of the tubes **39**.

(199) The path main body **119** extends straight without branching on the way in the axial direction of the screw main body **37**. The figures show, as an example, a state in which the path main body **119** extends parallel to the axial line **O1**. Both sides of the path main body **119** are closed in the axial direction.

(200) The entrance **117** is provided on one side of the path main body **119**, that is, a portion closer to the tip of the screw main body **37**. In this case, the entrance **117** may be opened in the outer circumferential surface of the screw main body **37** from an end face on the one side of the path main body **119**, or may be opened in the outer circumferential surface of the screw main body **37** from a portion closer to the end face on the one side of the path main body **119**, that is, a portion located short of the end face. The opening direction of the entrance **117** is not limited to those orthogonal to the axial line **O1**, but may be those crossing the axial line **O1**. In this case, entrances **117** may be provided by opening one side of the path main body **119** in directions.

(201) The exit **118** is provided on the other side (the opposite side to the one side) of the path main body **119**, that is, a portion closer to the basal end of the screw main body **37**. In this case, the exit **118** may be opened in the outer circumferential surface of the screw main body **37** from an end face on the other side of the path main body **119**, or may be opened in the outer circumferential surface of the screw main body **37** from a portion closer to the end face on the other side of the path main body **119**, that is, a portion located short of the end face. The opening direction of the exit **118** is not limited to those orthogonal to the axial line **O1**, but may be those crossing the axial line **O1**. In this case, exits **118** may be provided by opening one side of the path main body **119** in directions.

(202) The path main body **119** connecting the entrance **117** and the exit **118** has a length stretching over the tube **39** on which the first flight **110** is formed in each of the circulation portions **103**. In this case, the bore of the path main body **119** may be set to be smaller than those of the entrance **117** and the exit **118**, or may be set to be equal to them. In either case, the path sectional area defined by the bore of the path main body **119** is set to be much smaller than the annular sectional area in the radial direction of the above-described annular conveyance path **53**.

(203) In the present embodiment, if the screw **21** is disassembled by detaching the tubes **39** on which the flights **105**, **107**, **110**, **111**, and **112** are formed from the axis of rotation **38**, the tubes **39** on which at least parts of the flights **105**, **107**, **110**, **111**, and **112** are formed can also be referred to as the above-described screw elements.

(204) Thus, the screw main body **37** of the screw **21** can be formed by sequentially disposing the tubes **39** as the screw elements on the outer circumference of the axis of rotation **38**. Therefore, the conveyance portions **101** and the barrier portions **102** can be exchanged and rearranged in accordance with, for example, the degree of kneading of raw materials, and the exchange and the rearrangement can be easily performed.

(205) Moreover, by constricting the tubes **39** in the axial direction of the second axial portion **41** and firmly affixing the end faces of the adjacent tubes **39** to each other, the path main body **119** of each of the paths **115** is formed, and the entrance **117** and the exit **118** of each of the paths **115** are integrally connected through the path main body **119**. Thus, in order to form the paths **115** in the screw main body **37**, it suffices that each of the tubes **39** having a length much shorter than the total length of the screw main body **37** is processed. Thus, the paths **115** are easily processed and handled when being formed.

(206) As shown in FIG. **24**, the tube **39** on which the first flight **110** is formed is divided into two, such that the path main body **119** of the path **115** is divided. In one tube **39_t**, a lateral hole bored in the axial direction from a division surface **39_a** communicates with the exit **118**. In the other tube **39_p**, a lateral hole bored in the axial direction from a division surface **39_b** communicates with the entrance **117**. In this structure, the one continuous path **115**, both ends of which are opened in the

outer circumferential surface of the tube **39**, is formed by bringing the division surfaces **39a** and **39b** into contact with each other.

(207) As another path **115**, a path **115** may be formed to penetrate the tube **39** of the first flight **110** in the axial direction, for example, as shown in FIG. **25**. In this case, the entrance **117** and the exit **118** of the path **115** are opened in the inner surfaces of an entrance groove **120** and an exit groove **121** formed by cutting out parts of both the end faces in the axial direction of the tube **39** into the shape of depressions. According to this structure, even if the tube **39** is not divided, the one continuous path **115** can be formed simply by making a lateral hole penetrate the tube **39**.

(208) According to a continuous high shearing processing apparatus **1** having the above-described structure, a first extruder **2** preliminarily kneads resins. The resins melted by the kneading become raw materials having flowability, and are continuously supplied from the first extruder **2** to the conveyance path **53** through the supply port **34** of a second extruder **3**.

(209) As indicated by arrow D in FIG. **26**, the raw materials supplied to the second extruder **3** are introduced to the outer circumferential surface of the conveyance portion **101** positioned on the basal end side of the screw main body **37**. At this time, if the screw **21** rotates left-handed in an anticlockwise direction from the perspective of the basal end of the screw main body **37**, the flights **105** of the conveyance portions **101** continuously convey the raw materials toward the tip of the screw main body **37** as indicated by solid-line arrows in FIG. **26**.

(210) Then, the raw materials which have reached the circulation portions **103** are further continuously conveyed in the direction of the tip of the screw main body **37** by the first to third flights **110**, **111**, and **112** of the circulation portions **103** as indicated by solid-line arrows in FIG. **26** and FIG. **27**.

(211) In the meantime, shearing action, caused by a difference in speed between the flights **105**, **110**, **111**, and **112** rotating along the conveyance path **53** and the inner circumferential surface of the cylinder portion **33**, is imparted to the raw materials, and the raw materials are stirred by a subtle twist of the flights **105**, **110**, **111**, and **112**. As a result, the raw materials are kneaded thoroughly, and the dispersion of macromolecular components of the raw materials progresses.

(212) The raw materials, which have been subjected to shearing action, reach boundaries between the circulation portions **103** and the barrier portions **102** along the conveyance path **53**. In other words, the raw materials are fed to the boundaries between the circulation portions **103** and the barrier portions **102** by the first flights **110** disposed on the downstream side in the conveyance direction. On the other hand, when the screw **21** rotates left-handed, the flights **107** of the barrier portions **102** convey raw materials from the tip toward the basal end of the screw main body **37**.

(213) As a result, raw materials fed by the first flights **110** are stopped by the flights **107**. In other words, when the screw **21** rotates left-handed, the flights **107** of the barrier portions **102** limit the flow of raw materials fed by the first flights **110**.

(214) At this time, the pressure on the raw materials is increased at the boundaries between the circulation portions **103** and the barrier portions **102**. Specifically, FIG. **27** shows the filling rate of raw materials at the places in the conveyance path **53**, which correspond to the path **115**, with gradations. That is, in the conveyance path **53**, the filling rate of raw materials becomes greater as the tone becomes darker. As is clear from FIG. **27**, the filling rate of raw materials becomes greater as they approach the barrier portions **102** in the conveyance path **53** corresponding to the paths **115**, and just before the barrier portions **102**, the filling rate of raw materials is 100%.

(215) Thus, just before the barrier portions **102**, a “raw-material receiver R” in which the filling rate of raw materials is 100% is formed. In the raw-material receiver R, the flow of raw materials is stopped, and thus, the pressure on the raw materials is increased. As indicated by broken-line arrows in FIG. **26** and FIG. **27**, the raw materials, the pressure on which has been increased, continuously flow into the path main body **119** from the entrance **117**, and continuously flow through the path main body **119** from the tip toward the basal end of the screw main body **37**. At this time, the flow direction of raw materials in the path main body **119** is opposite to that of raw

materials fed by the flights **105**, **110**, **111**, and **112**.

(216) As described above, the path sectional area defined by the bore of the path main body **119** is much smaller than the annular sectional area of the conveyance path **53** in the radial direction of the cylinder portion **33**. From another point of view, a widening area based on the bore of the path main body **119** is much smaller than that of the annular conveyance path **53**. Therefore, raw materials are rapidly squeezed when flowing from the entrance **117** into the path main body **119**, and thus, extension action is imparted to the raw materials.

(217) Moreover, since the path sectional area is sufficiently smaller than the annular sectional area, raw materials collecting in the raw-material receiver R do not disappear. That is, some of the raw materials collecting in the raw-material receiver R continuously flow into the entrance **117**. In the meantime, new raw materials are fed toward the barrier portions **102** by the first flights **110**. As a result, the filling rate just before the barrier portions **102** in the raw-material receiver R is thereby kept at 100% all the time. At this time, even if the amount of raw materials conveyed by the first flights **110** somewhat changes, the change is absorbed by raw materials remaining in the raw-material receiver R. Raw materials can be thereby continuously and stably supplied to the paths **115**. Thus, in the paths **115**, extension action can be uninterruptedly and continuously imparted to the raw materials.

(218) The raw materials which have passed through the path main body **119** flow out of the exit **118** as indicated by solid-line arrows in FIG. **27**. The raw materials are thereby continuously returned to the outer circumferential surfaces of the circulation portions **103**. The returned raw materials are continuously conveyed toward the barrier portions **102** adjacent on the tip side of the screw main body **37** by the first flights **110**, and are subjected to shearing action again in the process of being conveyed.

(219) In this case, the portions on which the second flights **111** are formed can be given a backflow prevention function by setting the twist pitch of the second flights **111** smaller than that of the first flights **110**. This makes it possible to convey raw materials which have been returned from the exit **118** to the circulation portions **103** toward the barrier portions **102** without making them flow back.

(220) In the present embodiment, some of the raw materials conveyed toward the barrier portions **102** are continuously guided again from the entrance **117** to the paths **115**, and temporarily repeat circulation at the circulation portions **103**. The other raw materials conveyed toward the barrier portions **102** pass through the clearance between the apexes of the flights **107** of the barrier portions **102** and the inner circumferential surface of the cylinder portion **33**, and continuously flow into the adjacent circulation portions **103**.

(221) The barrier portions **102** and the circulation portions **103** are alternately arranged in the axial direction of the screw main body **37**, and the paths **115** provided at the positions corresponding to the first flights **110** of the circulation portions **103** are arranged with a space therebetween in the axial direction of the screw main body **37**. Thus, the raw materials supplied to the screw main body **37** from the supply port **34** are continuously conveyed from the basal end toward the tip of the screw main body **37** while being alternately and repeatedly subjected to shearing action and extension action. Thus, the degree of kneading of raw materials is increased, and the dispersion of macromolecular components of the raw materials is promoted.

(222) Then, the raw materials which have reached the tip of the screw main body **37** become sufficiently kneaded materials, and are continuously supplied to a third extruder **4** from a discharge port **36a**, and gaseous materials and other volatile components included in the kneaded materials are continuously removed from the kneaded materials.

(223) As described above, according to the fourth embodiment, the completely continuous production, not apparently continuous production, of kneaded materials is enabled. That is, resin preliminarily kneaded by the first extruder **2** continues being uninterruptedly supplied to the second extruder **3**, and thus, the flow of resin does not temporarily stagnate inside the first extruder **2**. Thus, temperature change, viscosity change, or phase change of the resin, caused when the kneaded

resin stagnates inside the first extruder **2**, can be prevented. As a result, raw materials having uniform quality all the time can be supplied to the second extruder **3** from the first extruder **2**.

(224) Furthermore, according to the fourth embodiment, shearing action and extension action can be alternately imparted to raw materials several times by the circulation portions **103** in which the paths **115** are formed. In this case, the number of times shearing action and extension action are imparted to raw materials can be further increased by disposing the circulation portions **103** in the axial direction.

(225) Moreover, according to the fourth embodiment, the paths **115** imparting extension action to raw materials extend in the axial direction of the screw main body **37** at positions eccentric to the axial line **O1**, which is the center of rotation of the screw main body **37**. Thus, the paths **115** revolve around the axial line **O1**. In other words, the tubular wall surfaces **116** defining the paths **115** revolve around the axial line **O1** without rotating on the axial line **O1**.

(226) Accordingly, when raw materials pass through the paths **115**, the raw materials are not actively stirred inside the paths **115**. Therefore, the raw materials passing through the paths **115** are hardly subjected to shearing action, and the raw materials passing through the paths **115** and returning to the outer circumferential surfaces of the conveyance portions **103** are mainly subjected to extension action. Therefore, also in the screw **21** of the fourth embodiment, places where shearing action is imparted to raw materials and places where extension action is imparted to raw materials can be clearly determined.

(227) Here, results of a high dispersion verification test performed on kneaded materials in the case where raw materials are kneaded while shearing action and extension action are alternately imparted to them by the above-described completely continuous production will be described.

(228) In the test, two kinds of materials, a polycarbonate (PC) resin and a polymethyl methacrylate (PMMA) resin, are supplied to the first extruder **2**, in which the effective length (L/D) of a kneading portion **12** with respect to the screw effective length (L/D) **50** is set at 7.9, and materials in a melted state are produced by preliminarily kneading them. In addition, the materials in a melted state are continuously supplied from the first extruder **2** to the second extruder **3** as raw materials of the second extruder **3**.

(229) In the test, the screw **21** is configured, such that the above-described circulation portions **103** are disposed at three places in the axial direction and raw materials pass through each of the paths **115**. Further, the specifications of the screw **21** are set as follows: the screw diameter is set at 36 mm, the screw effective length (L/D) is set at 16.7, the screw rotational rate is set at 2,500 rpm, the supply of raw materials is set at 10.0 kg/h, and the barrel set temperature is set at 240° C.

(230) Through the above-described test, intended transparent kneaded materials were continuously obtained.

Fifth Embodiment

(231) FIG. **28** shows a fifth embodiment. In the above-described first embodiment, the case where a first extruder (processor) **2** is formed as a twin screw kneader has been described. In the fifth embodiment, however, the case where the first extruder **2** is formed as a single screw extruder will be assumed instead.

(232) As shown in FIG. **28**, in the first extruder **2** according to the fifth embodiment, a barrel **6** comprises a cylinder portion **8** in which a single screw **7** is rotatably accommodated. As in the above-described first embodiment, the barrel **6** is provided with a supply port **9** through which, for example, pelletized materials can be supplied to the cylinder portion **8**, a heater (not shown in the figure) for melting resin, and a discharge port **6a** through which the melted resin can be discharged.

(233) The screw **7** can rotate on an axial line **O2**, and a spirally twisted flight **122** is formed on its outer circumferential surface. The flight **122** is configured to continuously convey resin supplied from the supply port **9** toward the discharge port **6a**. Thus, the flight **122** is twisted in the opposite direction to the rotation direction of the screw **7** from the perspective of the supply port **9** side. The figure shows, as an example, the flight **122** in the case where resin is conveyed by rotating the

screw **7** left-handed. In this case, the twist direction of the flight **122** is set to be clockwise as in the case of a right-handed screw.

(234) Moreover, a supply portion **P1**, a compression portion **P2**, and a conveyance portion **P3** are continuously formed on the outer circumferential surface of the screw **7** in order from the supply port **9** side toward the discharge port **6a**. The supply portion **P1** has a columnar shape, and a gap between its outer circumferential surface **7-P1** and the cylinder portion **8** is set wide. The conveyance portion **P3** has a columnar shape, and a gap between its outer circumferential surface **7-P3** and the cylinder portion **8** is set narrow. In other words, in the conveyance portion **P3**, the height of the flight **122** is set small by narrowing the gap between the outer circumferential surface **7-P3** and the cylinder portion **8**. The discharge stability of the discharge port **6a** is thereby improved. The compression portion **P2** has a shape widening from the supply portion **P1** toward the conveyance portion **P3**, and a gap between its outer circumferential surface **7-P2** and the cylinder portion **8** is set to become continuously narrower from the supply portion **P1** toward the conveyance portion **P3**.

(235) Here, in a state in which the screw **7** is rotated left-handed, pelletized resin supplied from the supply port **9** to the cylinder portion **8** is conveyed by the flight **122** in the order of the supply portion **P1**, the compression portion **P2**, and the conveyance portion **P3**, and then discharged from the discharge port **6a**. In the supply portion **P1**, the resin has a low temperature and is in a solid state. In the compression portion **P2**, the resin is mainly compressed by the gap that continuously becomes narrower while being heated by the heater. In the conveyance portion **P3**, the resin forms melted and mixed raw materials. Then, raw materials discharged from the discharge port **6a** of the barrel **6** are continuously supplied to a second extruder **3** as indicated by arrow **A** in FIG. **1**.

(236) As described above, according to the fifth embodiment, even if the first extruder **2** is formed as a single screw extruder, raw materials of optimum viscosity for a kneading process performed by the second extruder **3** can be produced as in the above-described case of the twin screw kneader according to the first embodiment. The workload of the second extruder **3** can be thereby reduced.

(237) For example, assuming that shearing action and extension action are alternately imparted to already preliminarily kneaded materials, that is, materials pelletized by incorporating a filler (additives) into resin, the materials can be kneaded without causing deterioration of the physical properties of the additives or cutting of fibers by using a single screw extruder.

(238) In addition, if additives are added to raw materials, the physical properties of the additives may be deteriorated or the additives may be decomposed by high-speed rotation in the second extruder **3** when the additives are introduced to the first extruder **2** or the second extruder **3**. In this case, if a third extruder **4** is formed as a twin screw extruder, the third extruder **4** can incorporate (knead) the additives into the raw materials as well as performing deaeration.

Sixth Embodiment

(239) FIG. **29** shows a sixth embodiment. The sixth embodiment differs from the first embodiment in the structure for imparting extension action to raw materials. The other structures of a screw **21** are basically the same as those of the first embodiment.

(240) As shown in FIG. **29**, a pair of grooves **131a** and **131b** is formed in the inner circumferential surface of a tube **39**. The grooves **131a** and **131b** extend in the axial direction of a screw main body **37**, and are remote from each other in the radial direction of the screw main body **37**. Moreover, the grooves **131a** and **131b** are opened in the inner circumferential surface of the tube **39**.

(241) When the tube **39** is inserted on a second axial portion **41** of an axis of rotation **38**, opening ends of the grooves **131a** and **131b** are closed by the outer circumferential surface of the second axial portion **41**. Thus, the grooves **131a** and **131b** define paths **132** which impart extension action to raw materials in cooperation with the outer circumferential surface of the second axial portion **41**. In the present embodiment, the paths **132** are positioned at a boundary between the axis of rotation **38** and the tube **39**.

(242) According to the sixth embodiment, the paths **132** are provided at positions eccentric to an

axial line O1 of the axis of rotation 38 inside the screw main body 37. Accordingly, as in the first embodiment, the paths 132 are shifted from the axial line O1, and revolve around the axial line O1 when the screw main body 37 rotates.

(243) In the sixth embodiment, the paths 132 are formed inside the screw main body 37 when the tube 39 is inserted on the second axial portion 41 of the axis of rotation 38. Since the grooves 131a and 131b defining the paths 132 are opened in the inner circumferential surface of the tube 39, the grooves 131a and 131b can be easily formed.

(244) Therefore, for example, even if it becomes necessary to change the sectional shape of the paths 132, it can be easily changed.

Seventh Embodiment

(245) FIG. 30 shows a seventh embodiment. The seventh embodiment differs from the sixth embodiment in the structure for imparting extension action to raw materials. The other structures of a screw 21 are basically the same as those of the sixth embodiment.

(246) As shown in FIG. 30, a pair of grooves 141a and 141b are formed in the outer circumferential surface of a second axial portion 41 of an axis of rotation 38. The grooves 141a and 141b extend in the axial direction of the second axial portion 41, and are remote from each other in the radial direction of the second axial portion 41. Moreover, the grooves 141a and 141b are opened in the outer circumferential surface of the second axial portion 41.

(247) When a tube 39 is inserted on the second axial portion 41 of the axis of rotation 38, opening ends of the grooves 141a and 141b are closed by the inner circumferential surface of the tube 39. Thus, the grooves 141a and 141b define paths 142 which impart extension action to raw materials in cooperation with the inner circumferential surface of the tube 39. In the present embodiment, the paths 142 are positioned at a boundary between the axis of rotation 38 and the tube 39.

(248) According to the seventh embodiment, the paths 142 are provided at positions eccentric to an axial line O1 of the axis of rotation 38 inside a screw main body 37. Accordingly, as in the sixth embodiment, the paths 142 are shifted from the axial line O1, and revolve around the axial line O1 when the screw main body 37 rotates.

(249) In the seventh embodiment, the paths 142 are formed inside the screw main body 37 when the tube 39 is inserted on the second axial portion 41 of the axis of rotation 38. Since the grooves 141a and 141b defining the paths 142 are opened in the outer circumferential surface of the axis of rotation 38, the grooves 141a and 141b can be easily formed.

(250) Therefore, for example, even if it becomes necessary to change the sectional shape of the paths 142, it can be easily changed.

Eighth Embodiment

(251) FIG. 31 shows an eighth embodiment. The eighth embodiment differs from the first embodiment in the structure for imparting extension action to raw materials. The other structures of a screw 21 are basically the same as those of the first embodiment.

(252) As shown in FIG. 31, depressions 151a and 151b are formed in the tip surfaces of keys 45a and 45b projecting from the outer circumferential surface of a second axial portion 41. The depressions 151a and 151b extend in the axial direction of the second axial portion 41, and are opened in the tip surfaces of the keys 45a and 45b. When the keys 45a and 45b are fitted into keyways 49a and 49b of a tube 39, opening ends of the depressions 151a and 151b are closed by the inner peripheral surfaces of the keyways 49a and 49b.

(253) Thus, the depressions 151a and 151b define paths 152 which impart extension action to raw materials in cooperation with the inner peripheral surfaces of the keyways 49a and 49b. In the present embodiment, the paths 152 are positioned at boundaries between the keys 45a and 45b and the tube 39.

(254) According to the eighth embodiment, the paths 152 are provided at positions eccentric to an axial line O1 of an axis of rotation 38 inside a screw main body 37. Accordingly, as in the first embodiment, the paths 152 are shifted from the axial line O1, and revolve around the axial line O1

when the screw main body **37** rotates.

(255) In the eighth embodiment, the paths **152** are formed inside the screw main body **37**, when the keys **45a** and **45b** of the axis of rotation **38** are fitted into the keyways **49a** and **49b** of the tube **39**. Since the depressions **151a** and **151b** defining the paths **152** are opened in the tip surfaces of the keys **45a** and **45b**, the depressions **151a** and **151b** can be easily formed.

(256) Therefore, for example, even if it becomes necessary to change the sectional shape of the paths **152**, it can be easily changed.

(257) In the eighth embodiment, the paths **152** may be defined by providing other depressions extending in the axial direction of the second axial portion **41** in the inner circumferential surfaces of the keyways **49a** and **49b**, and fitting the other depressions to the depressions **151a** and **151b**.

Ninth Embodiment

(258) FIG. **32** shows a ninth embodiment. The ninth embodiment differs from the first embodiment in the structure of a screw **21** and the structure for imparting extension action to raw materials.

(259) As shown in FIG. **32**, the screw **21** comprises a solid screw main body **161**. The screw main body **161** is composed of a straight axial member **162**. The axial member **162** has an axial line **O1** coaxially penetrating its central portion, and is coaxially accommodated in a cylinder portion **33** of a barrel **20**.

(260) Moreover, the axial member **162** comprises an outer circumferential surface **162a** continuing in the circumferential direction, and the outer circumferential surface **162a** faces the inner circumferential surface of the cylinder portion **33** of the barrel **20**. A flight (not shown in the figure) conveying raw materials is formed on the outer circumferential surface **162a** of the axial member **162**.

(261) Moreover, a pair of paths **164** imparting extension action to raw materials is formed inside the axial member **162**. The paths **164** extend in the axial direction of the axial member **162**, and are disposed to be parallel to each other with the axial line **O1** sandwiched therebetween. Therefore, the paths **164** are provided at positions eccentric to the axial line **O1** of the axial member **162** inside the screw main body **161**. Accordingly, as in the first embodiment, the paths **164** are shifted from the axial line **O1**, and revolve around the axial line **O1** when the screw main body **161** rotates.

(262) The paths **164** imparting extension action to raw materials can be formed inside the screw main body **161** even if the screw main body **161** is composed of the rodlike member **162**.

Therefore, the screw main body is not limited to a combination of an axis of rotation and a tube.

Other Embodiments

(263) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiment described herein may be made without departing from the spirit of the invention.

(264) For example, the sectional shape of a path imparting extension action to raw materials is not limited to a circular hole. The path may be composed of, for example, a hole having an elliptical or polygonal sectional shape, and the sectional shape of the path is not particularly limited.

(265) In addition, in each of the above-described embodiments, the case where the screw **21** rotates left-handed in an anticlockwise direction when the screw main body is viewed from the direction of the basal end of the axis of rotation **38** has been described as example. The present invention, however, is not limited to this. For example, the screw **21** may be rotated right-handed in a clockwise direction from the perspective of the basal end side of the screw **21**.

(266) In this case, for example, in the first embodiment, it suffices that the flights **56** of the conveyance portions **54** of the screw **21** are twisted right-handed as in the case of a right-handed screw to convey raw materials from the tip toward the basal end of the screw main body **37**.

Similarly, it suffices that the flights **57** of the barrier portions **55** are twisted left-handed as in the case of a left-handed screw to convey raw materials from the basal end toward the tip of the screw

main body **37**.

(267) Moreover, the barrier portions of the screw main body are not restricted to being composed of spirally twisted flights. For example, the barrier portions may be composed of annular major diameter portions having outer circumferential surfaces continuing in the circumferential direction of the screw main body. It is preferable that the major diameter portions have widths in the axial direction of the screw main body, and have smooth annular shapes without depressions, cutouts, etc., in their outer circumferential surfaces.

(268) In addition, the third extruder **4** which removes gaseous components included in kneaded materials extruded from the second extruder **3** is not limited to a single screw extruder, and may be a twin screw extruder. If the third extruder **4** is formed as a twin screw extruder, it suffices that two vented screws **23** identical to the vented screw **23** shown in FIG. **4** are arranged in parallel, and the respective flights **29** are engaged with each other in a state of being out of phase by 90°. Surface renewal of kneaded materials can be promoted by rotating the two screws **23** in the same direction. Thus, the efficiency of drawing and removal of gaseous components included in the kneaded materials can be improved. The kneaded materials, from which gaseous components are drawn and removed, are continuously discharged from the discharge port **28** of the head portion **27** to the outside of the high shearing processing apparatus **1**.

(269) It suffices that the continuous high shearing processing apparatus according to the present invention comprises at least a first extruder preliminarily kneading raw materials and a second extruder thoroughly kneading the raw materials. A third extruder removing gaseous materials and other volatile components may be omitted. If the third extruder is omitted, it is preferable that at least one vent-port removing gaseous materials and volatile components from raw materials in the process of being kneaded be provided in a middle portion of the second extruder.

(270) Moreover, as the first extruder (processor) **2**, not only the above-described twin screw kneader (see FIG. **2** and FIG. **3**) and single screw extruder (see FIG. **28**), but also various kneaders, for example, a multi-screw extruder, a Banbury mixer, a kneader, and an open roll can be used.

REFERENCE SIGNS LIST

(271) **2**: First extruder (processor) **3**: Second extruder **4**: Third extruder (deaerator) **20**: Barrel **21**: Screw **34**: Supply port **36a**: Discharge port **37**, **161**: Screw main body **54**, **81**, **101**: Conveyance portion **56**, **57**, **58**, **84**, **86**, **105**, **107**, **110**, **111**, **112**, **122**: Flight **60**, **88**, **115**, **132**, **142**, **152**, **164**: Path **O1**, **O2**: Axial line

Claims

1. A kneading method comprising: continuously melting and mixing materials in a processor to form melted materials; and using the melted materials as raw materials and continuously discharging kneaded materials produced by kneading the raw materials with a screw in an extruder, wherein the screw comprises a screw main body configured to rotate on a straight axial line in a conveyance direction of the raw materials such that tubes are coaxially joined to an axis of rotation whereby each of the tubes and the axis of the rotation are integrally assembled to form the screw, in the screw main body, a conveyance portion configured to convey the raw materials, a barrier portion configured to limit conveyance of the raw materials, and a plurality of paths through which raw materials flow are provided at places in a direction of the axial line, and the plurality of paths are arranged in a circumferential direction of the screw main body at intervals, each of the plurality of paths is provided inside each respective one of the tubes, and each of the plurality of paths comprises an entrance, an exit, and path main body communicating between the entrance and the exit, the entrance is opened in an outer circumferential surface of the tube in the conveyance portion to cause the raw materials, the conveyance of which is limited by the barrier portion to increase pressure on the raw materials, to flow in the entrance, each of plurality of paths is respectively configured to allow the raw materials to flow in from the entrance, pass through the

path main body, and flow out of the exit, the exit is opened in the outer circumferential surface of the tube at a position remote from the entrance in a direction of the axial line, in the extruder, the raw materials conveyed along an outer circumferential surface of the screw main body flow through the paths and then return to an outer circumferential surface of the screw, while the kneaded materials are continuously discharged, conveyance of the raw materials conveyed along the outer circumferential surface of the screw main body is limited by the barrier portion provided on the screw main body to increase pressure on the raw materials, and the raw materials, the pressure on which is increased by the barrier portion, are made to flow into the paths from the outer circumferential surface of the screw main body.

2. The kneading method of claim 1, wherein in a deaerator connected to the extruder, a gaseous component is drawn and removed from the kneaded materials and continuously discharged from the extruder.
